

Machine Learning

CMPSCI 589

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Linear Regression

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& Computer Sciences



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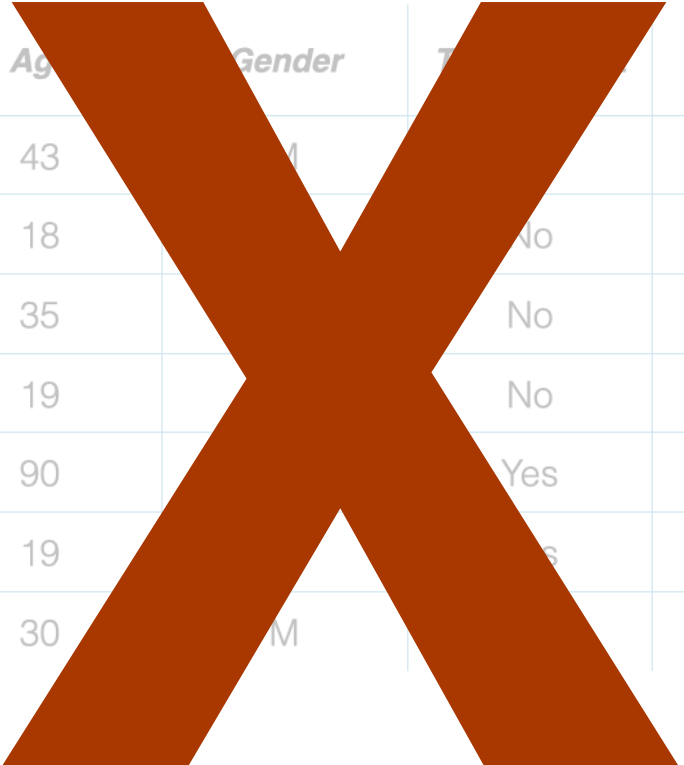
Machine Learning

- Classification algorithms: predict a class/label (discrete)

<i>Name</i>	<i>Age</i>	<i>Gender</i>	<i>TrafficTicket</i>	Class: High-Risk Driver
John	43	M	Yes	High Risk
Peter	18	M	No	High Risk
Anna	35	F	No	Low Risk
Paula	19	F	No	High Risk
Mark	90	M	Yes	High Risk
Marisa	19	F	Yes	High Risk
Bob	30	M	No	Low Risk

Machine Learning

- **Classification algorithms:** predict a class/label (discrete)
- **Regression algorithms:** predict a real number (i.e., models a function)



Name	Age	Gender	Traffic Violations	Class: High-Risk Driver
John	43	M	No	High Risk
Peter	18	M	No	High Risk
Anna	35	F	No	Low Risk
Paula	19	F	No	High Risk
Mark	90	M	Yes	High Risk
Marisa	19	F	Yes	High Risk
Bob	30	M	No	Low Risk

	#friends	funny score	Likes
Post1	40	8	28
Post2	36	10	28
Post3	20	6	16
Post4	56	4	31
Post5	58	0	29
Post6	46	10	33

Linear Regression

- Predict number of **Likes** an Instagram post/picture might get
- Based on
 - how many friends were online
 - how funny the user believes the meme is (0-10)



	#friends	funny score	Likes
Post1	40	8	28
Post2	36	10	28
Post3	20	6	16
Post4	56	4	31
Post5	58	0	29
Post6	46	10	33

(<http://qr.ae/TbcC8T>)

Linear Regression

	#friends	funny score	Likes	
Post1	40	8	28	28
Post2	36	10	28	28
Post3	20	6	16	16
Post4	56	4	31	32
Post5	58	0	29	29
Post6	46	10	33	33



It seems like
there is a *pattern* here!

$$\text{Likes} = (0.5 * \text{\#friends}) + (1 * \text{funny_score})$$

Machine learning algorithm will learn these "weights"

Linear Regression

	#friends	funny score	Likes	
Post1	40	8	28	28
Post2	36	10	28	28
Post3	20	6	16	16
Post4	56	4	31	32
Post5	58	0	29	29
Post6	46	10	33	33



It seems like
there is a *pattern* here!

$$\text{Likes} = (0.5 * \text{\#friends}) + (1 * \text{funny_score})$$

Linear Regression

Regression Problems

- Regression algorithms: predict a real number (i.e., models a function)

	x =LSD Dosage	y = Exam Score
Student1	1.17	78.93
Student2	2.97	58.20
Student3	3.26	67.47
Student4	4.69	37.47
Student5	5.83	45.65
Student6	6.00	32.92
Student7	6.41	29.97

*"Correlation of Performance Test Scores
with Tissue Concentration of
Lysergic Acid Diethylamide in Human Subjects".*

Wagner, Agahajanian, and Bing (1968)

Regression Problems

- Regression algorithms: predict a real number (i.e., models a function)

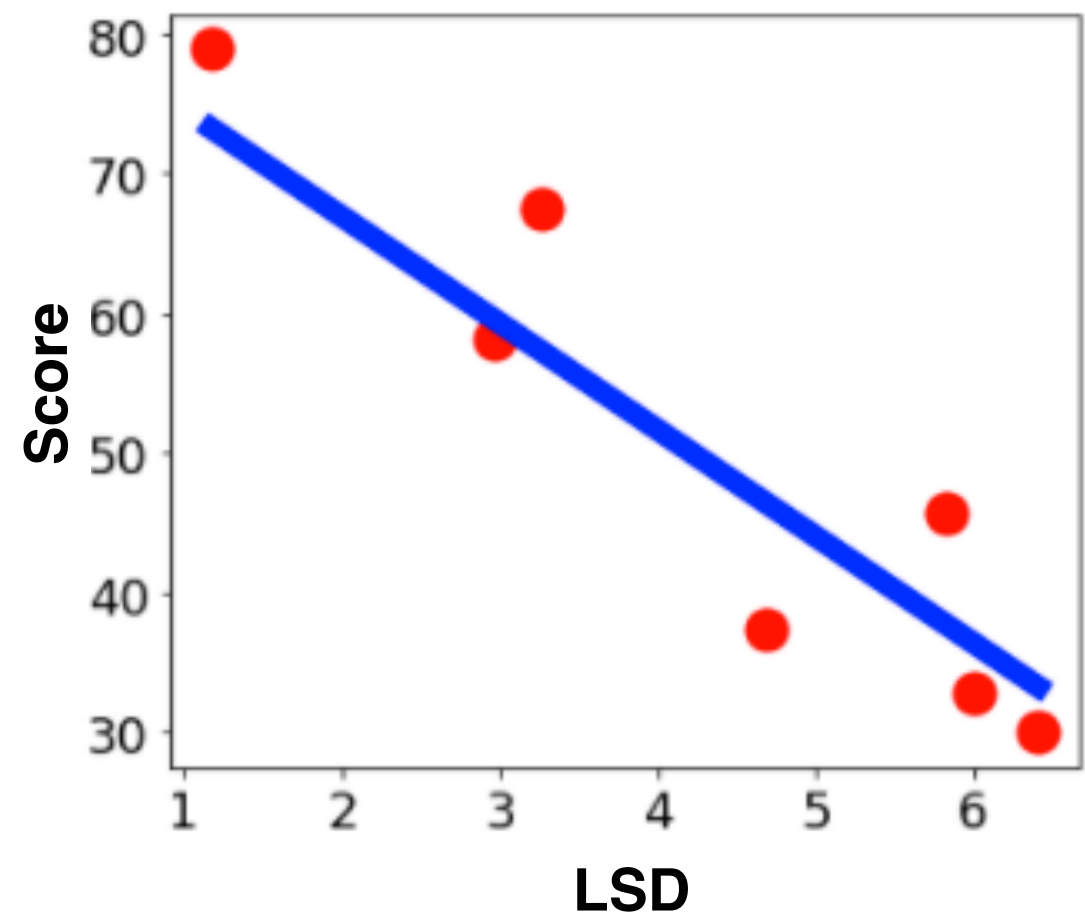
	x =LSD Dosage	y = Exam Score
Student1	1.17	78.93
Student2	2.97	58.20
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Student4	4.69	37.47
Student5	5.83	45.65
Student6	6.00	32.92
Student7	6.41	29.97

$f(\text{LSD_Dosage}) \rightarrow \text{Exam_Score}$

$f(x) \rightarrow y$

Hypothesis:
 $f(x) = \theta_0 + \theta_1 x$

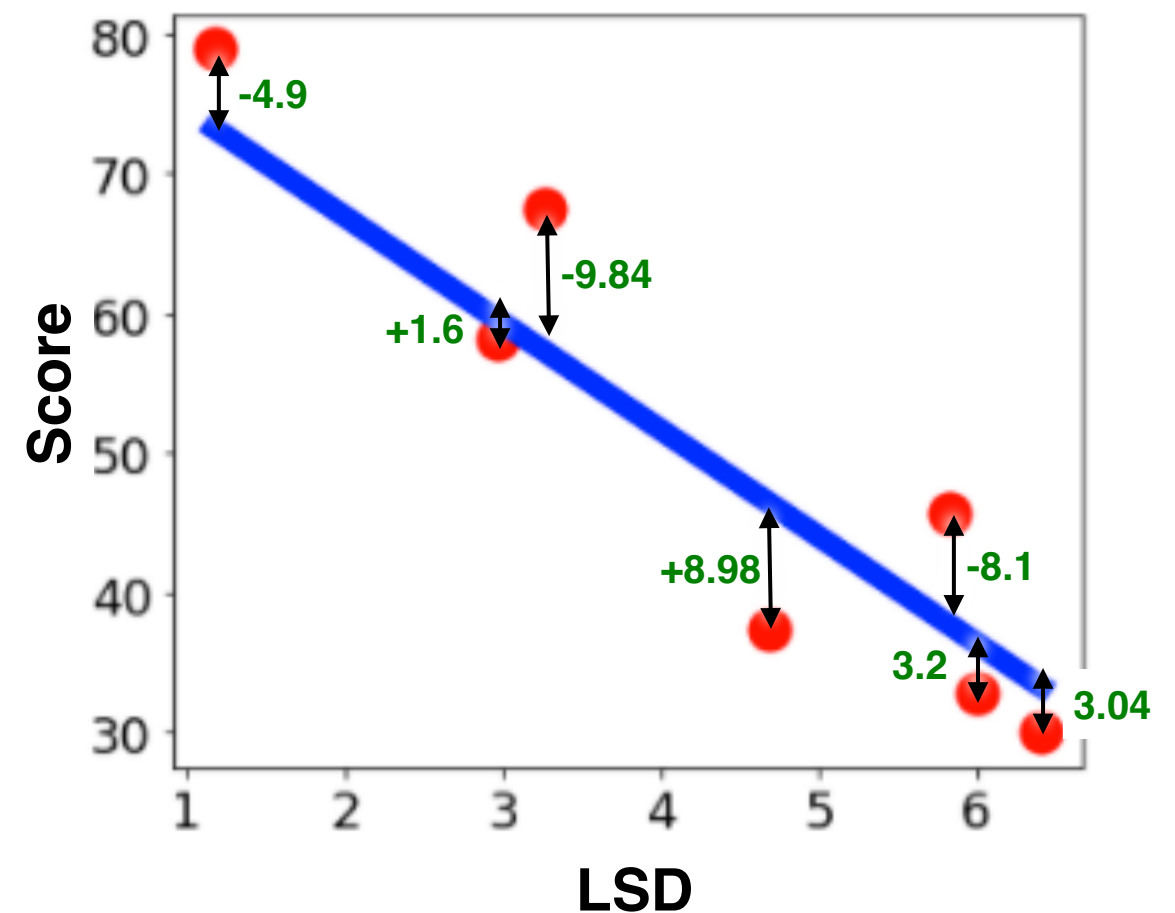
Assume $\theta_0 = 83.089$
 $\theta_1 = -7.812$



Regression Problems

- Regression algorithms: predict a real number (i.e., models a function)

	x =LSD Dosage	y = Exam Score	f(x) = Predicted Score
Student1	1.17	78.93	73.94
Student2	2.97	58.20	59.88
Student3	3.26	67.47	57.62
Student4	4.69	37.47	46.45
Student5	5.83	45.65	37.54
Student6	6.00	32.92	36.21
Student7	6.41	29.97	33.01



$f(\text{LSD_Dosage}) \rightarrow \text{Exam_Score}$

$f(x) \rightarrow y$

Hypothesis:
 $f(x) = \theta_0 + \theta_1 x$

Assume $\theta_0 = 83.089$
 $\theta_1 = -7.812$

How "wrong" is this model?

Regression Problems

- Regression algorithms: predict a real number (i.e., models a function)

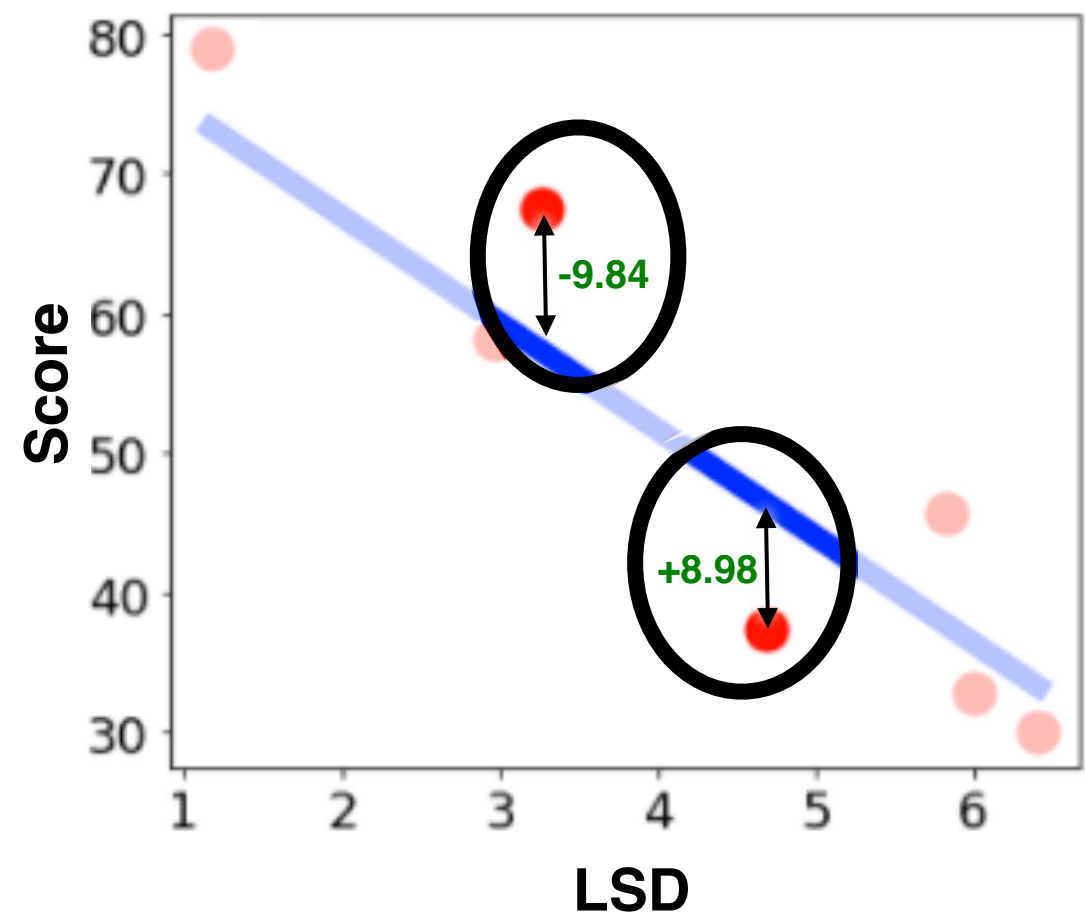
	x =LSD Dosage	y = Exam Score	f(x) = Predicted Score
Student1	1.17	78.93	73.94
Student2	2.97	58.20	59.88
Student3	3.26	67.47	57.62
Student4	4.69	37.47	46.45
Student5	5.83	45.65	37.54
Student6	6.00	32.92	36.21
Student7	6.41	29.97	33.01

$f(\text{LSD_Dosage}) \rightarrow \text{Exam_Score}$

$f(x) \rightarrow y$

Hypothesis:
 $f(x) = \theta_0 + \theta_1 x$

Assume $\theta_0 = 83.089$
 $\theta_1 = -7.812$



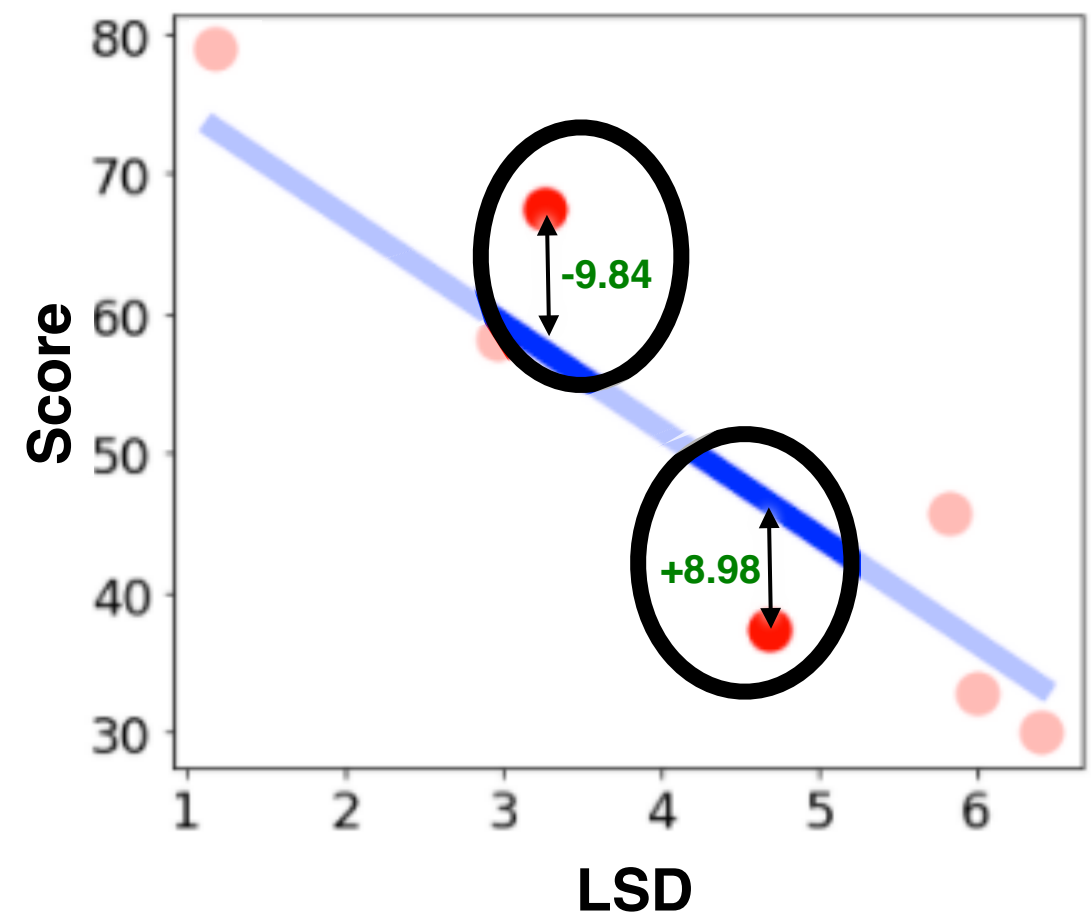
**How “wrong”
is this model?**

**Let’s try to
sum the errors**

Regression Problems

- Regression algorithms: predict a real number (i.e., models a function)

	x =LSD Dosage	y = Exam Score	f(x) = Predicted Score
Student1	1.17	78.93	73.94
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$f(\text{LSD_Dosage}) \rightarrow \text{Exam_Score}$

$f(x) \rightarrow y$

Hypothesis:
 $f(x) = \theta_0 + \theta_1 x$

Assume $\theta_0 = 83.089$
 $\theta_1 = -7.812$

**How “wrong”
 is this model?**

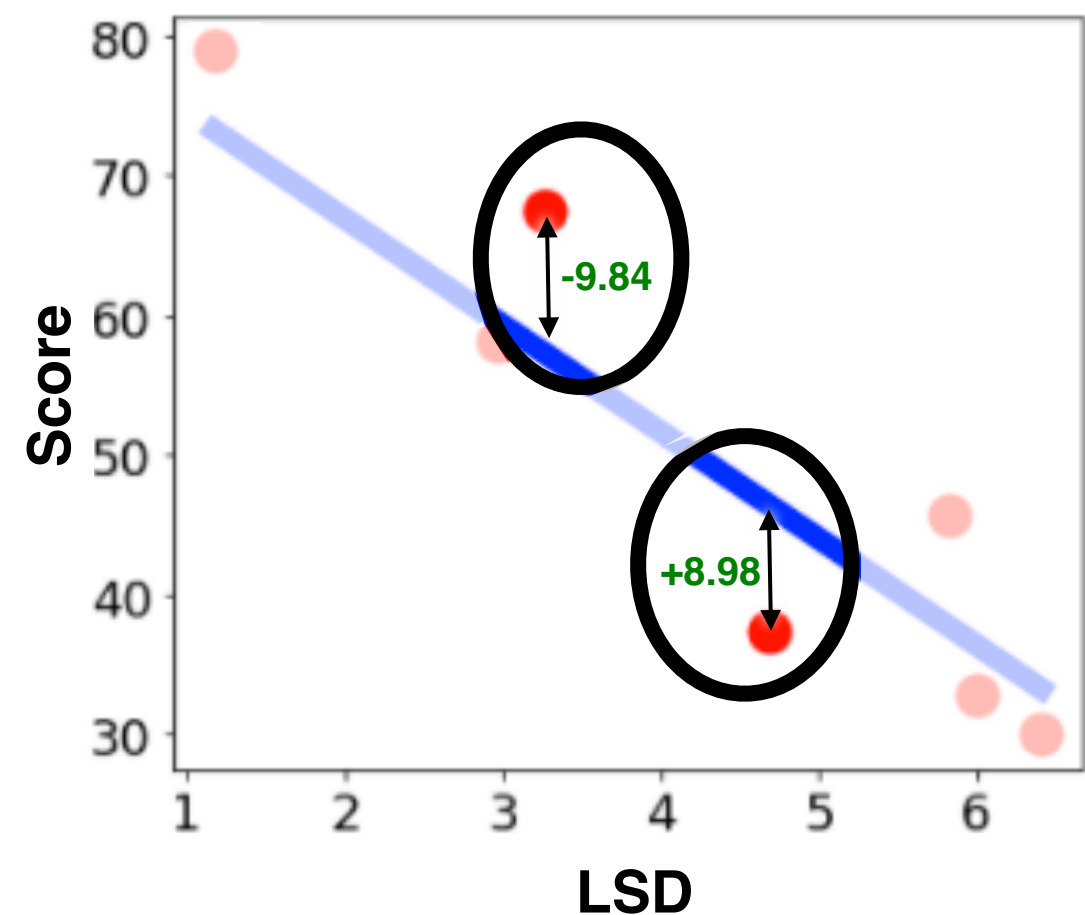
Sum of errors
 $-9.84 + 8.98 = 0.86$

**Positive and negative errors
 cancel out!**

Regression Problems

- Regression algorithms: predict a real number (i.e., models a function)

	x =LSD Dosage	y = Exam Score	f(x) = Predicted Score
Student1	1.17	78.93	73.94
Student2	2.97	58.20	59.88
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Student5	5.83	45.65	37.54
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Student7	6.41	29.97	33.01



$f(\text{LSD_Dosage}) \rightarrow \text{Exam_Score}$

$f(x) \rightarrow y$

Hypothesis:
 $f(x) = \theta_0 + \theta_1 x$

Assume $\theta_0 = 83.089$
 $\theta_1 = -7.812$

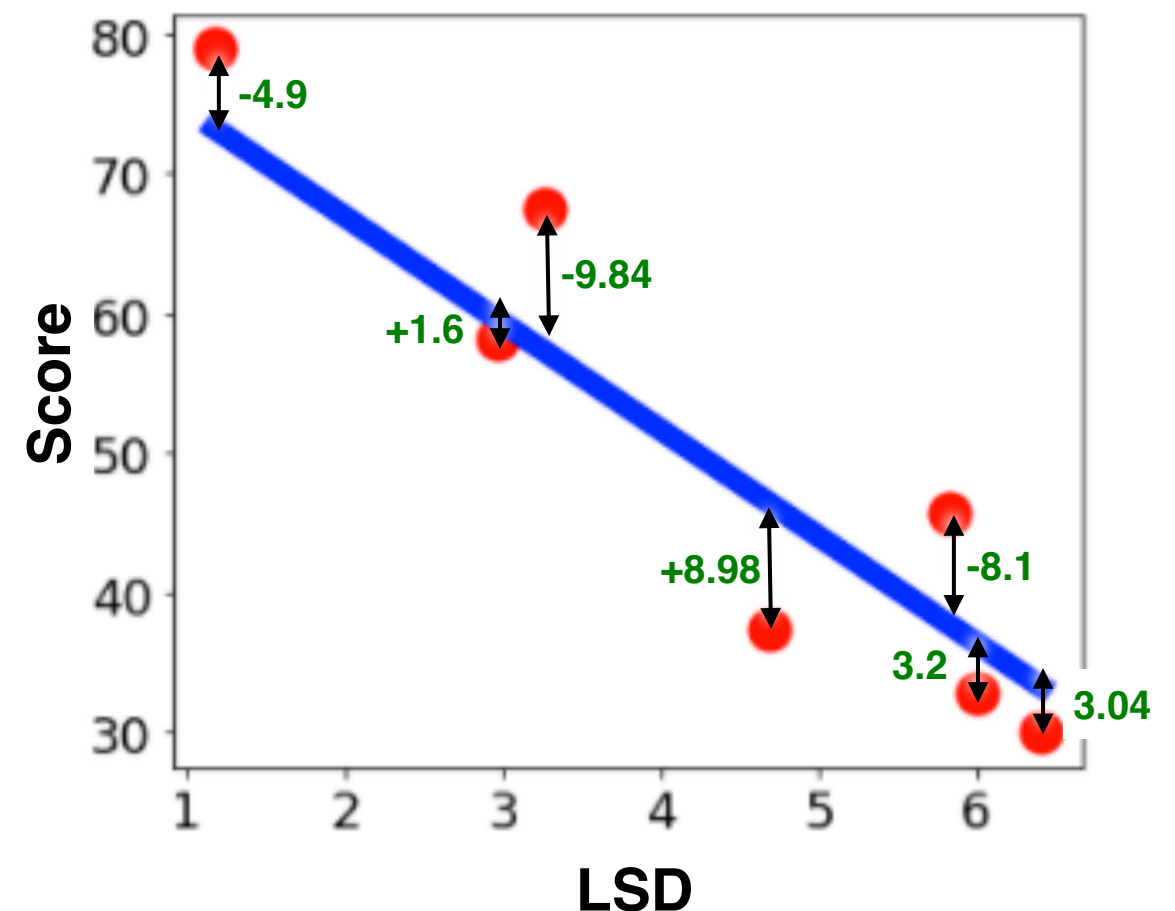
How “wrong”
is this model?

Sum of squared errors:
 $-9.84^2 + 8.98^2 = 177.47$

Regression Problems

- Regression algorithms: predict a real number (i.e., models a function)

	x =LSD Dosage	y = Exam Score	f(x) = Predicted Score
Student1	1.17	78.93	73.94
Student2	2.97	58.20	59.88
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$f(\text{LSD_Dosage}) \rightarrow \text{Exam_Score}$

$f(x) \rightarrow y$

Assume $\theta_0 = 83.089$
 $\theta_1 = -7.812$

How “wrong”
is this model?

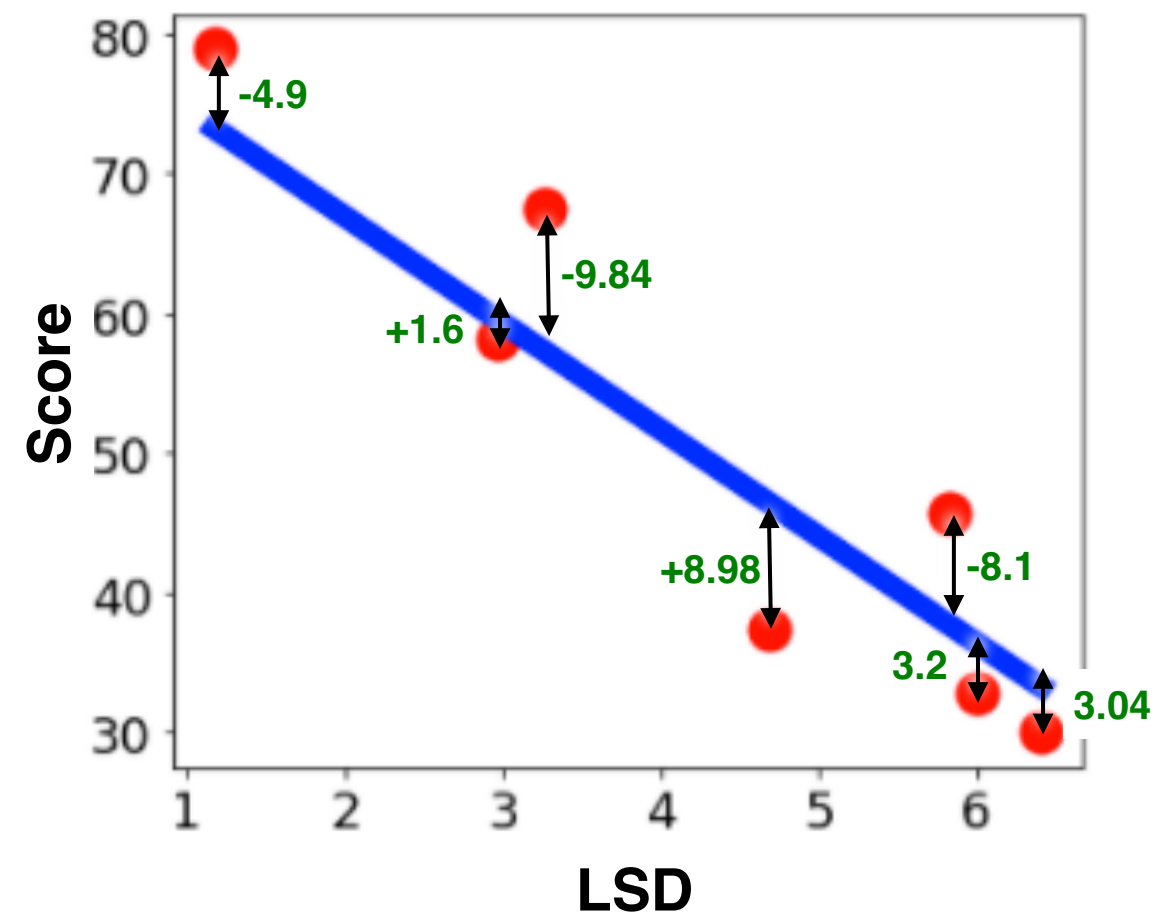
Sum of squared errors:

$$(-4.9)^2 + (1.6)^2 + (-9.84)^2 + (8.98)^2 + (-8.1)^2 + (3.2)^2 + (3.04)^2$$

Regression Problems

- Regression algorithms: predict a real number (i.e., models a function)

	x =LSD Dosage	y = Exam Score	f(x) = Predicted Score
Student1	1.17	78.93	73.94
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$f(\text{LSD_Dosage}) \rightarrow \text{Exam_Score}$

$f(x) \rightarrow y$

Assume $\theta_0 = 83.089$
 $\theta_1 = -7.812$

Mean squared error:

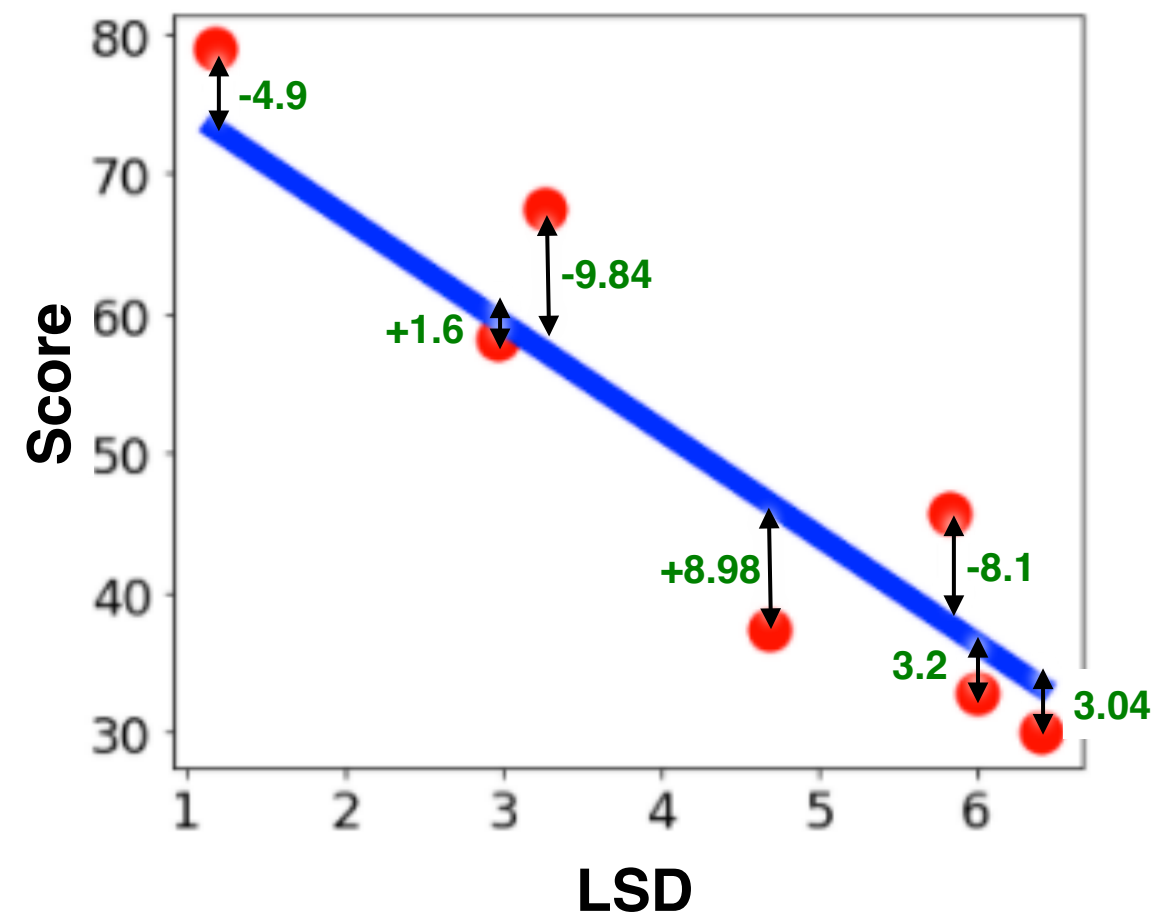
$$(1/7) \times \left[(-4.9)^2 + (1.6)^2 + (-9.84)^2 + (8.98)^2 + (-8.1)^2 + (3.2)^2 + (3.04)^2 \right] = 41.30$$

How “wrong”
 is this model?

Regression Problems

- Regression algorithms: predict a real number (i.e., models a function)

	x =LSD Dosage	y = Exam Score	f(x) = Predicted Score
Student1	1.17	78.93	73.94
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$f(\text{LSD_Dosage}) \rightarrow \text{Exam_Score}$
 $f(x) \rightarrow y$

Mean squared error:

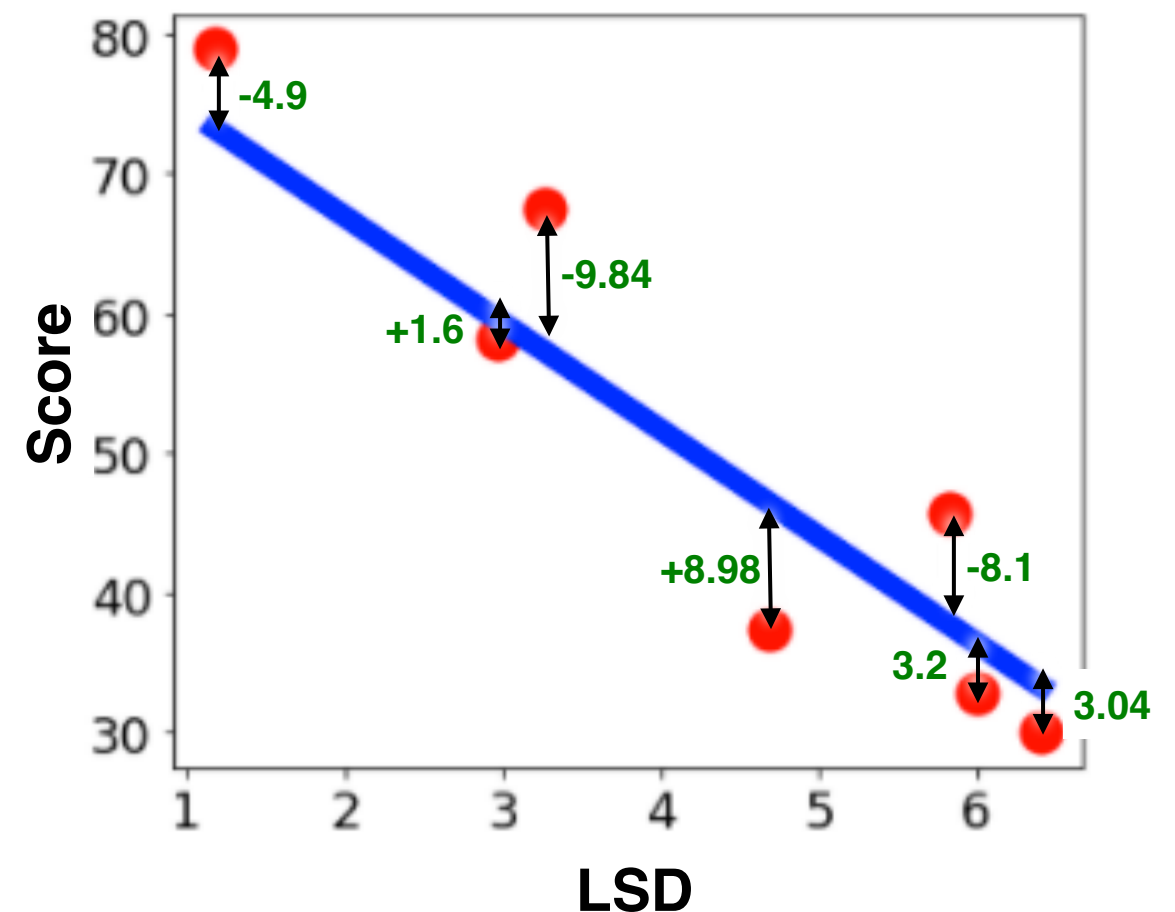
$$J(\theta_0, \theta_1) = \frac{1}{n} \times \left[(-4.9)^2 + (1.6)^2 + (-9.84)^2 + (8.98)^2 + (-8.1)^2 + (3.2)^2 + (3.04)^2 \right] = 41.30$$

$$= \sum_{i=1}^n (f_{\theta}(x^{(i)}) - y^{(i)})^2$$

Regression Problems

- Regression algorithms: predict a real number (i.e., models a function)

	x =LSD Dosage	y = Exam Score	f(x) = Predicted Score
Student1	1.17	78.93	73.94
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Student7	6.41	29.97	33.01



$f(\text{LSD_Dosage}) \rightarrow \text{Exam_Score}$
 $f(x) \rightarrow y$

$$J(\theta_0, \theta_1) = \frac{1}{n} \times \left[\sum_{i=1}^n \left(\underbrace{f_{\theta}(x^{(i)})}_{\text{predicted output given input } x^{(i)}} - \underbrace{y^{(i)}}_{\text{correct output given input } x^{(i)}} \right)^2 \right]$$

where $f_{\theta}(x) = \theta_0 + \theta_1 x$ is our hypothesis

predicted output
given input $x^{(i)}$

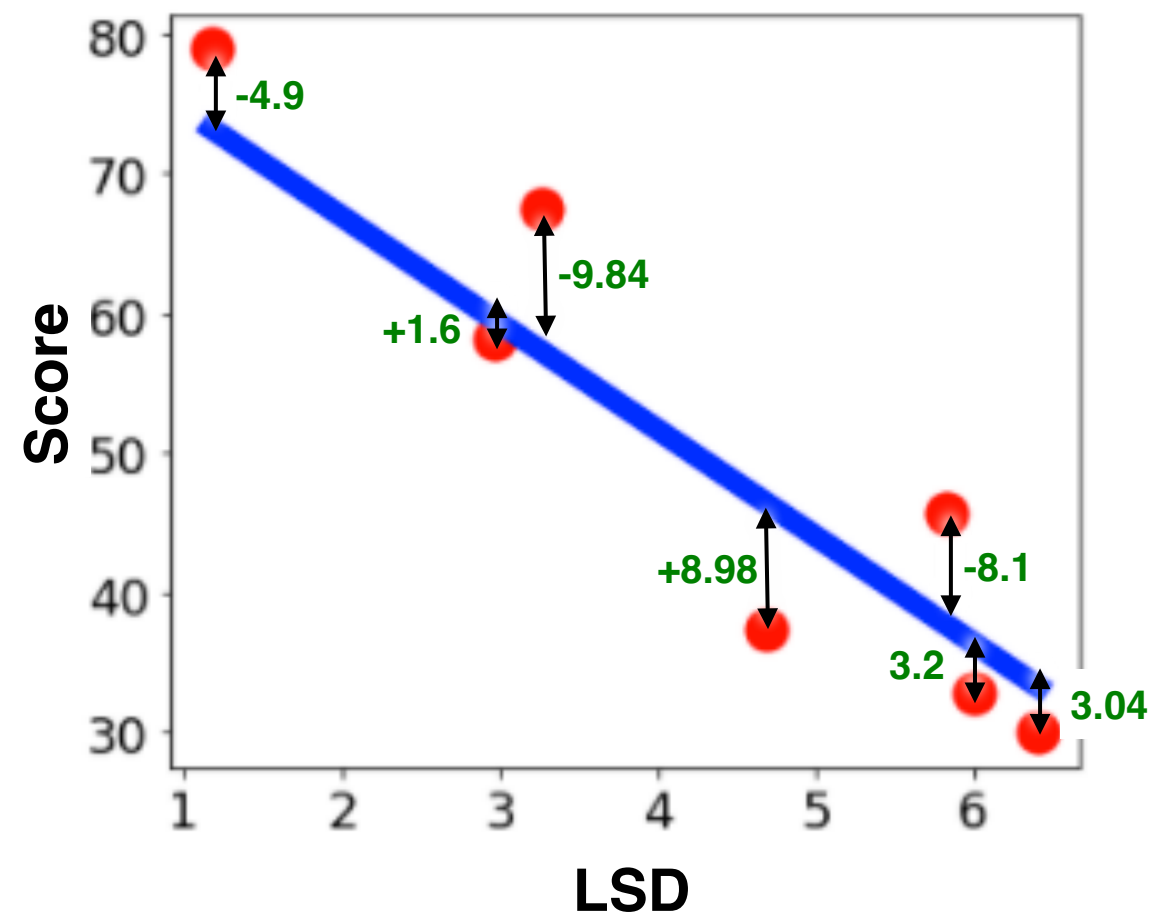
correct output
given input $x^{(i)}$

Regression Problems

- Regression algorithms: predict a real number (i.e., models a function)

	x =LSD Dosage	y = Exam Score	f(x) = Predicted Score
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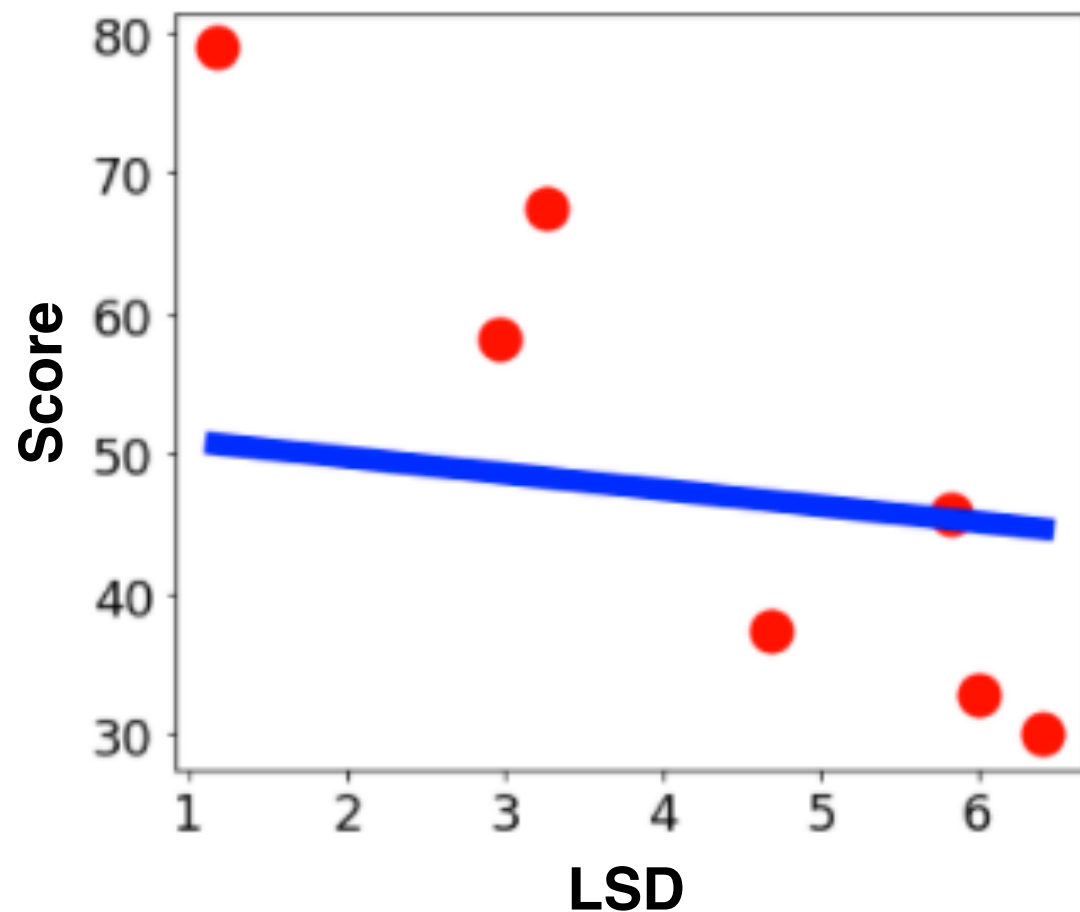
$f(\text{LSD_Dosage}) \rightarrow \text{Exam_Score}$
 $f(x) \rightarrow y$



$$J(\theta_0, \theta_1) = \frac{1}{2n} \sum_{i=1}^n (f_{\theta}(x^{(i)}) - y^{(i)})^2 \quad \text{where } f_{\theta}(x) = \theta_0 + \theta_1 x \quad \text{is our hypothesis}$$

We wish to find values for θ_0 and θ_1 that minimize the Mean Squared Error, $J(\theta_0, \theta_1)$

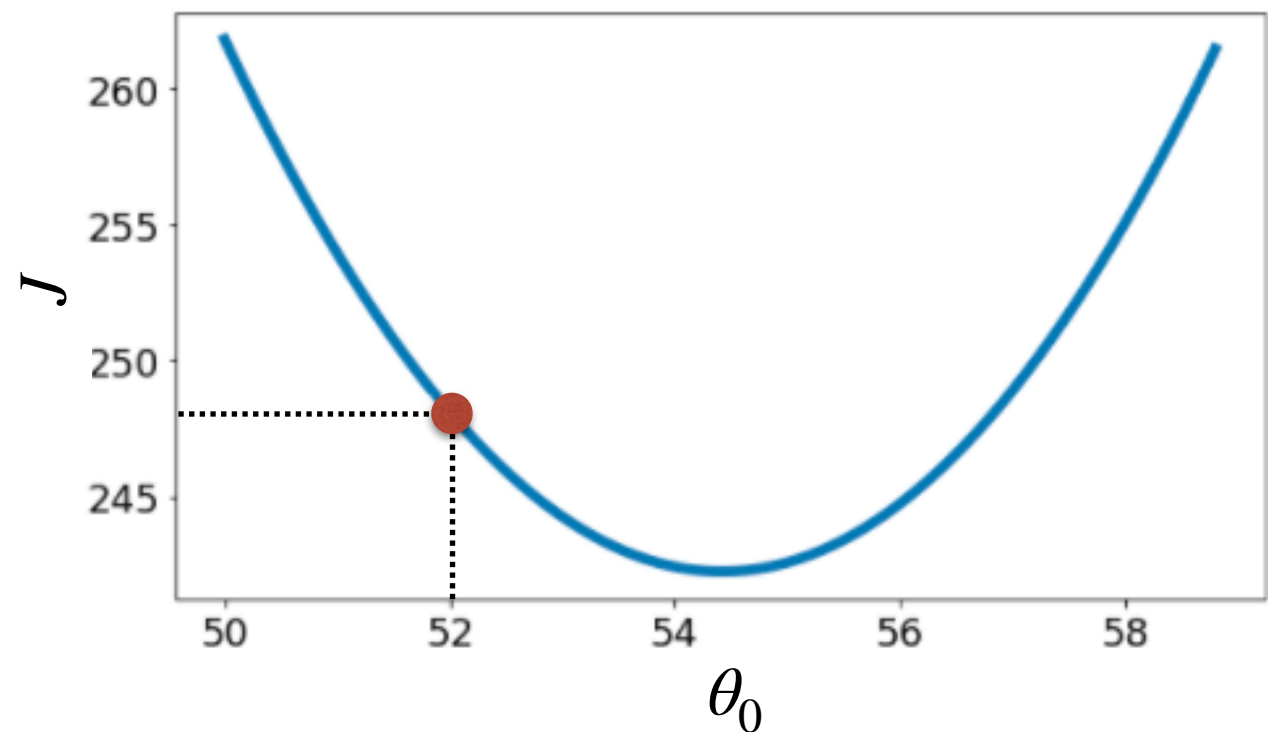
Regression Problems



Hypothesis:
 $f(x) = \theta_0 + \theta_1 x$

Assume $\theta_0 = 52.0$
 $\theta_1 = -1.0$

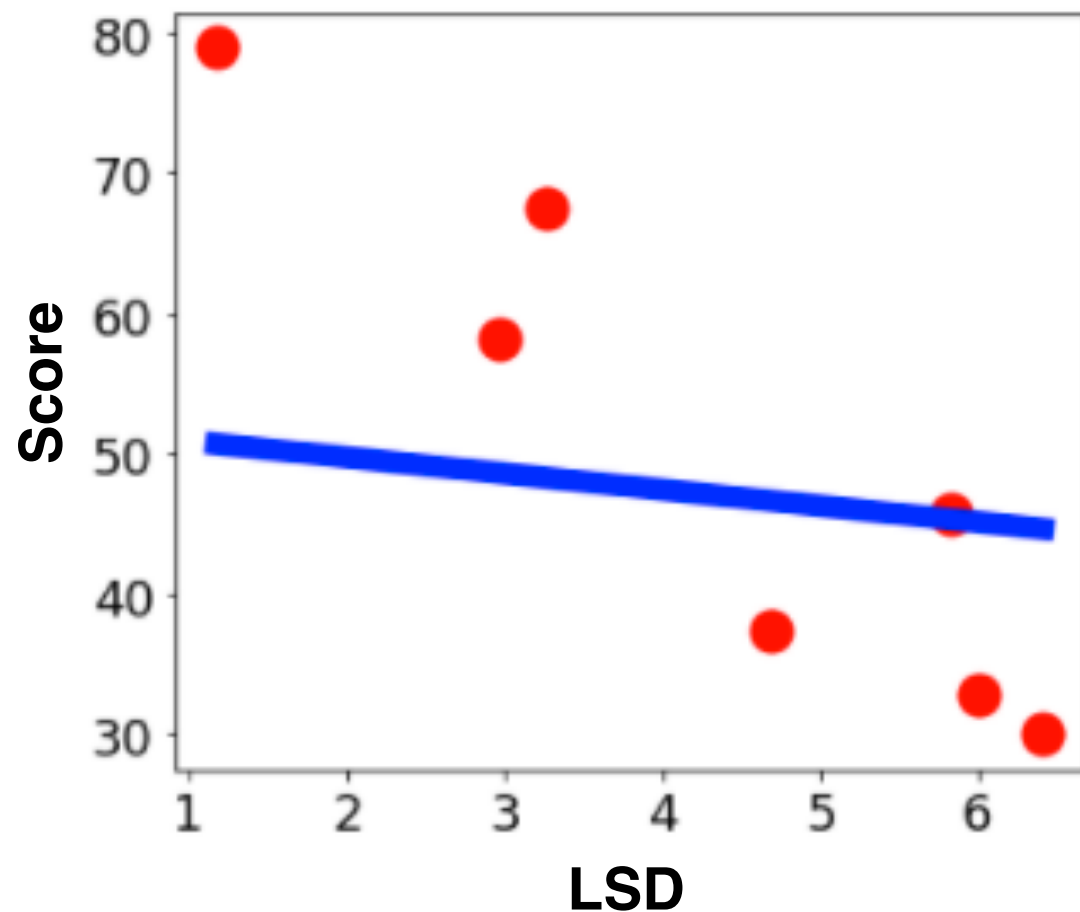
Let's keep $\theta_1 = -1.0$ fixed.
What happens with J as we change θ_0 ?



How should we update
 θ_0 to decrease the error J ?

Make θ_0 more positive

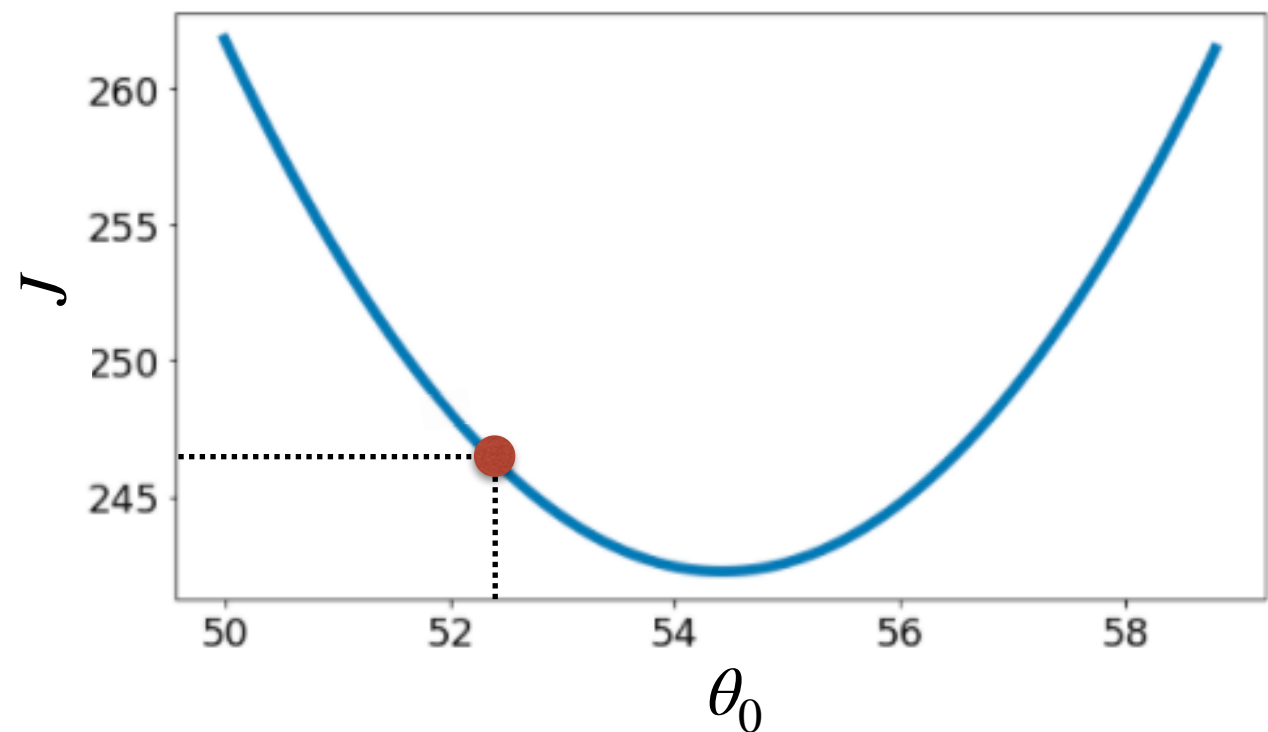
Regression Problems



Hypothesis:
 $f(x) = \theta_0 + \theta_1 x$

Assume $\theta_0 = 52.0$
 $\theta_1 = -1.0$

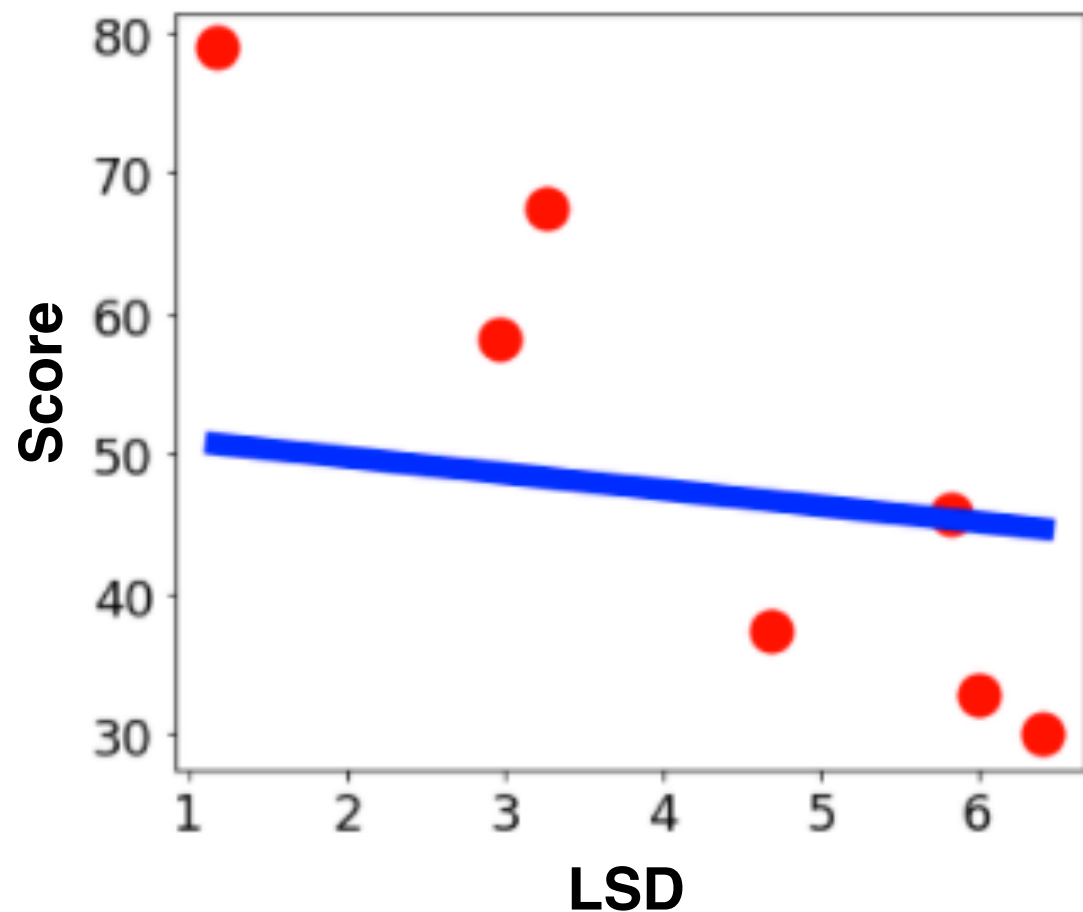
Let's keep $\theta_1 = -1.0$ fixed.
What happens with J as we change θ_0 ?



How should we update
 θ_0 to decrease the error J ?

Make θ_0 more positive

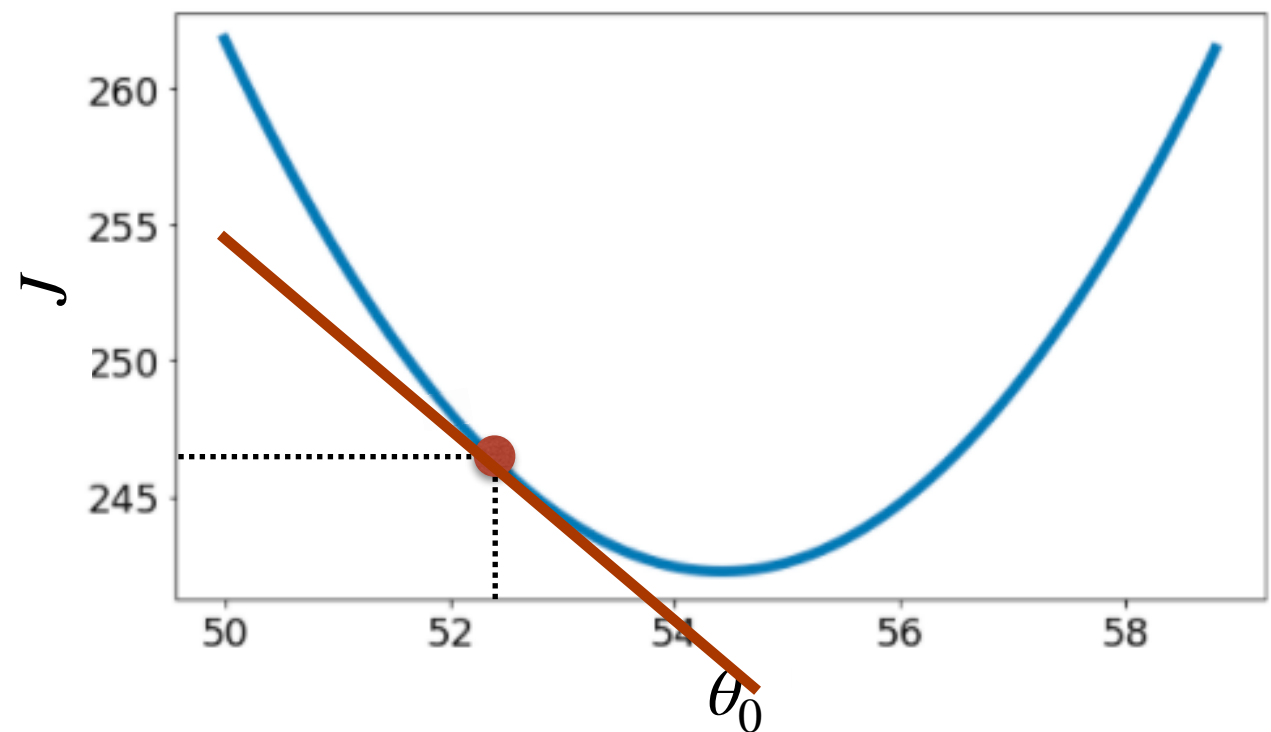
Regression Problems



Hypothesis:
 $f(x) = \theta_0 + \theta_1 x$

Assume $\theta_0 = 52.0$
 $\theta_1 = -1.0$

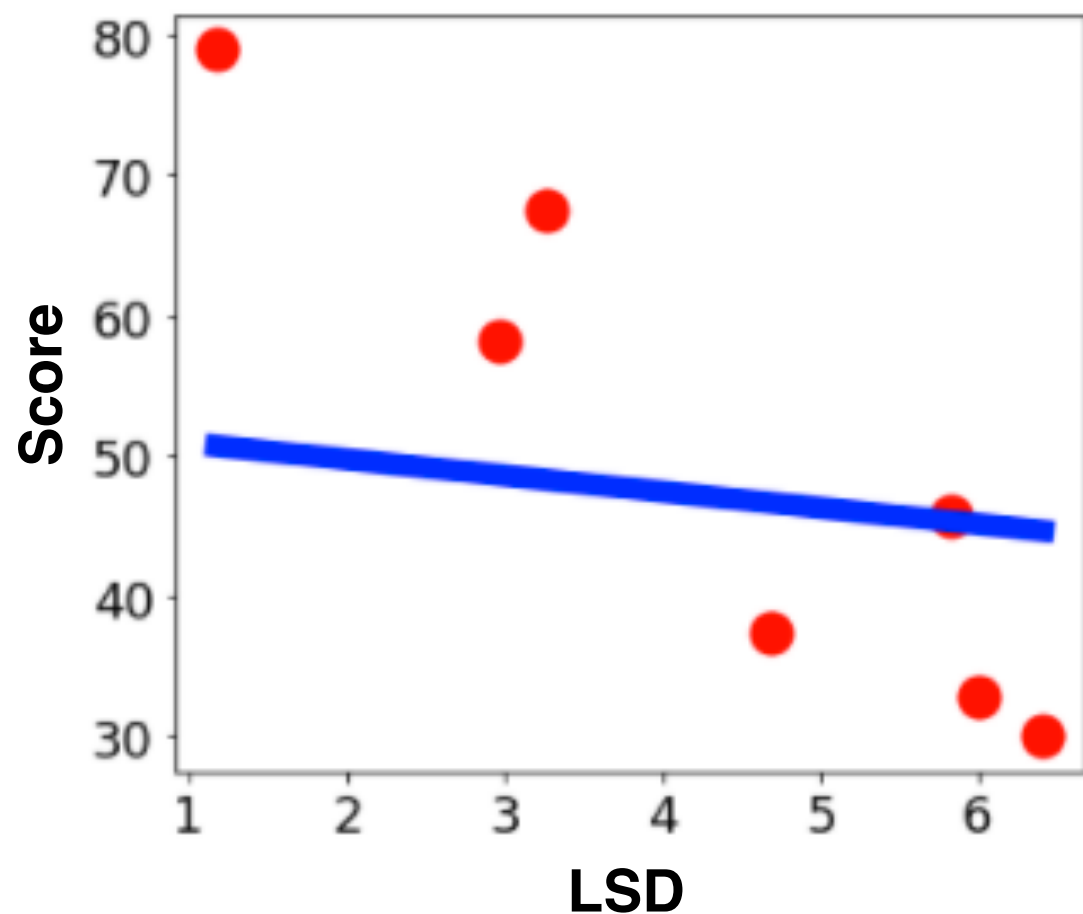
Let's keep $\theta_1 = -1.0$ fixed.
What happens with J as we change θ_0 ?



Make θ_0 more positive

Derivative of the J function with respect to θ_0
Slope of the function J at that point

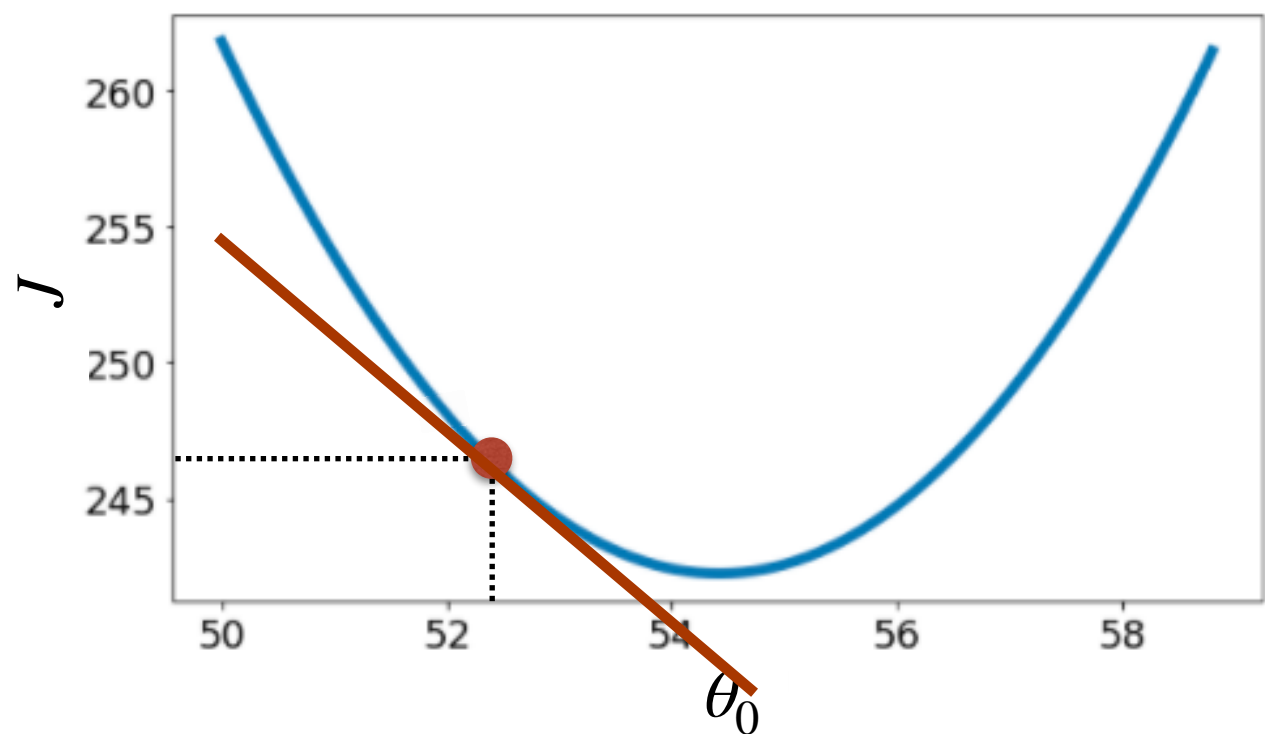
Regression Problems



Hypothesis:
 $f(x) = \theta_0 + \theta_1 x$

Assume $\theta_0 = 52.0$
 $\theta_1 = -1.0$

Let's keep $\theta_1 = -1.0$ fixed.
What happens with J as we change θ_0 ?



Make θ_0 more positive

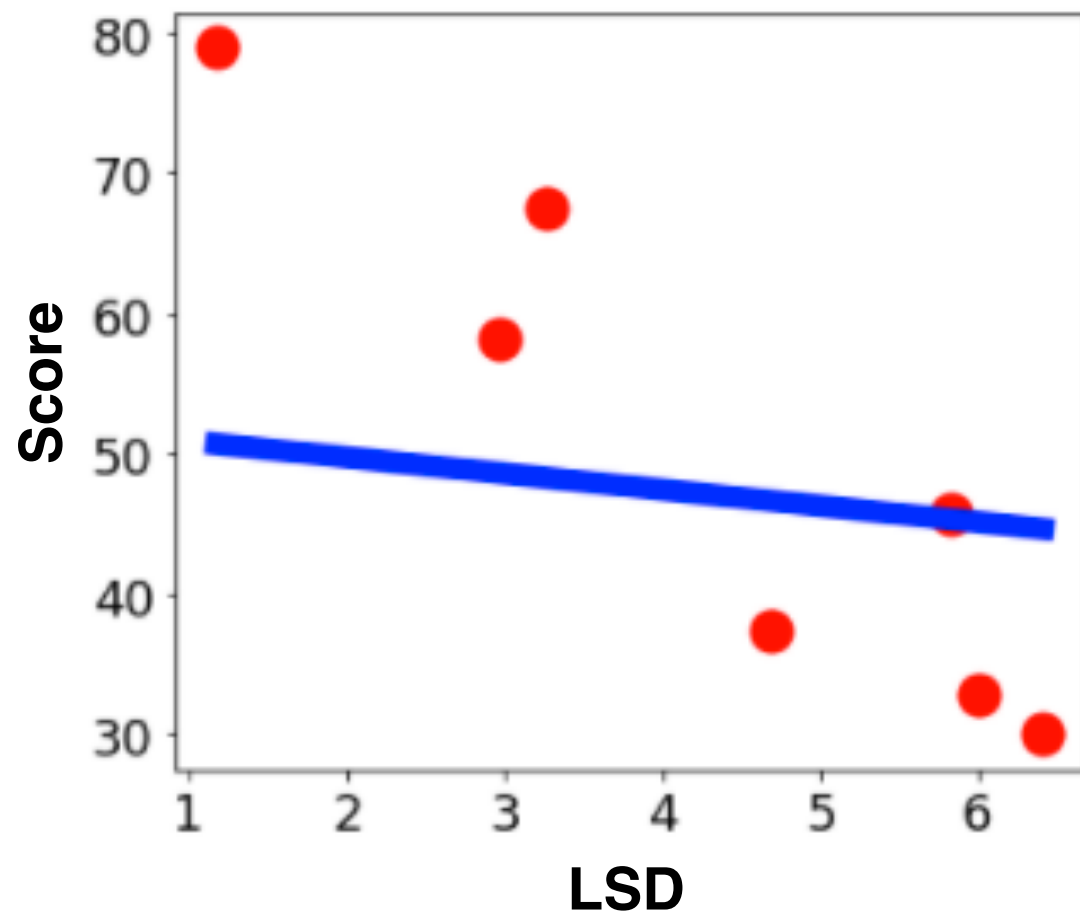
Derivative of the J function with respect to θ_0
 $= -15.22$ (negative derivative/slope)

→ if we make θ_0 more positive, J decreases

→ if we make θ_0 more negative, J increases

→ if derivative is negative, we should make θ_0 more positive

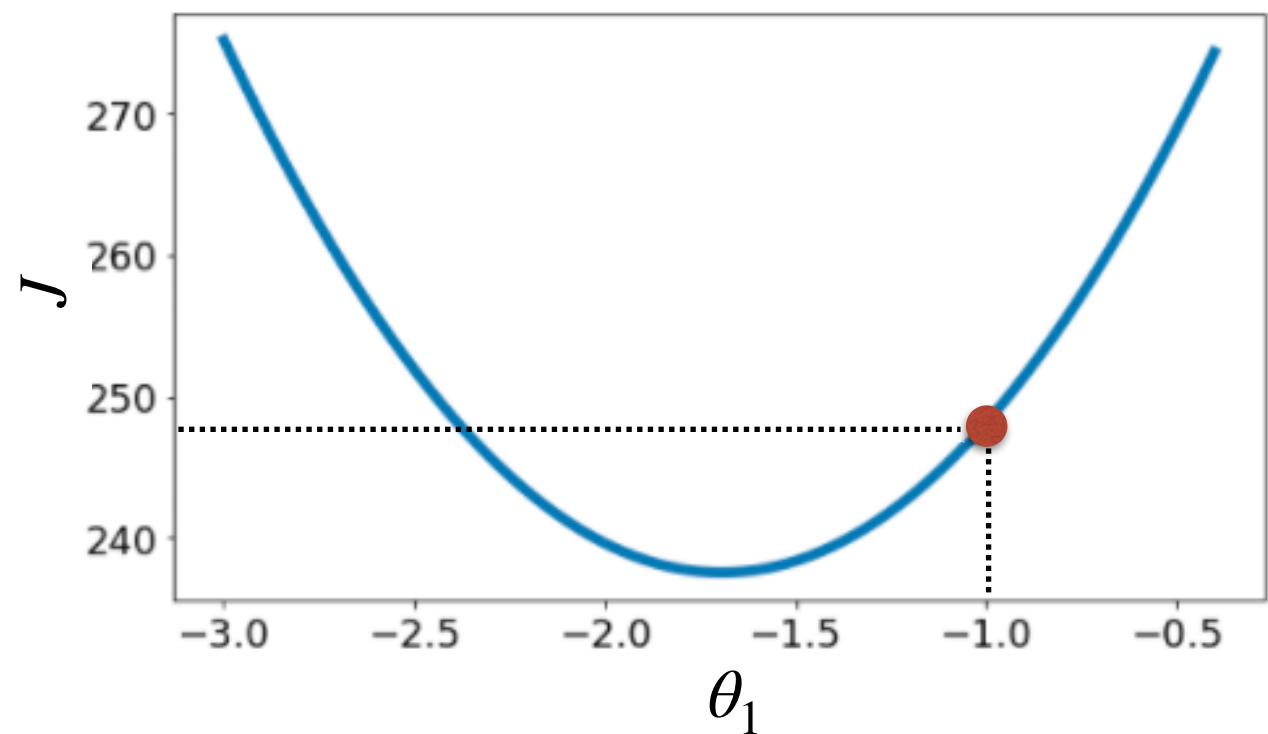
Regression Problems



Hypothesis:
 $f(x) = \theta_0 + \theta_1 x$

Assume $\theta_0 = 52.0$
 $\theta_1 = -1.0$

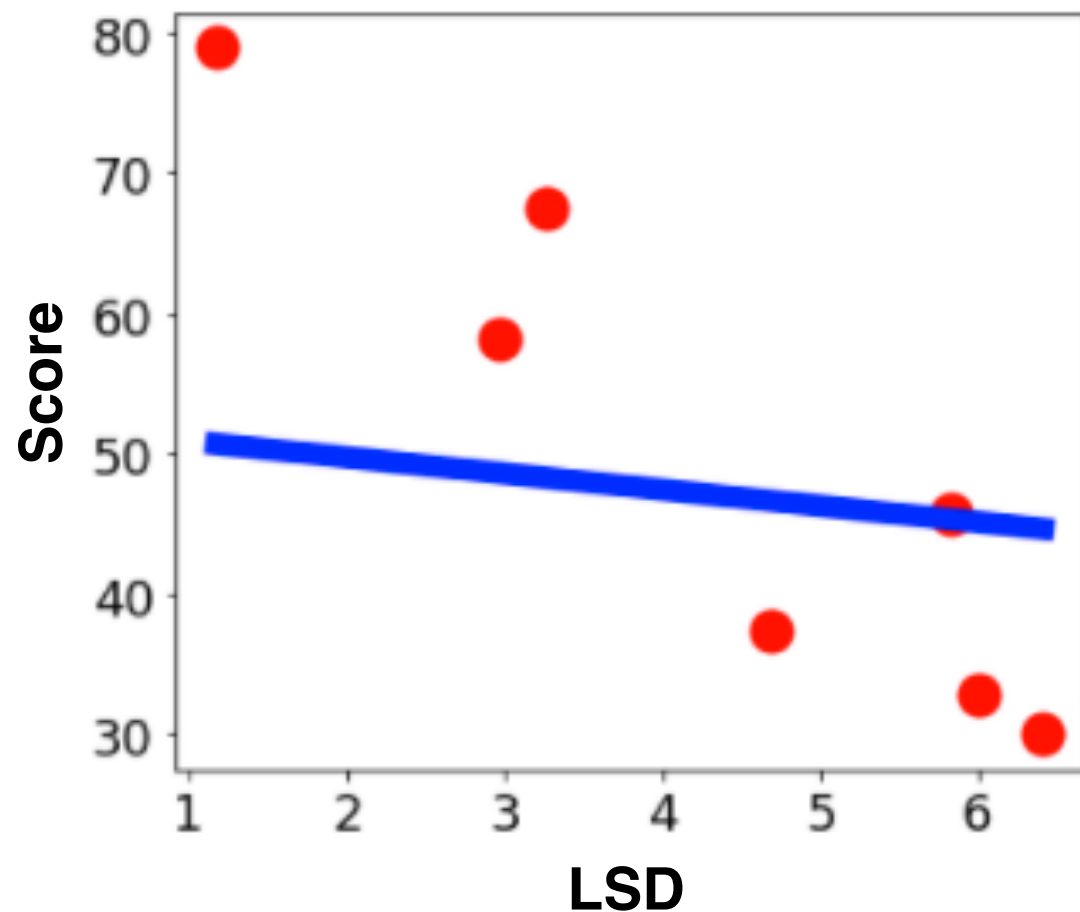
Let's keep $\theta_0 = 52.0$ fixed.
What happens with J as we change θ_1 ?



How should we update
 θ_1 to decrease the error J ?

Make θ_1 more negative

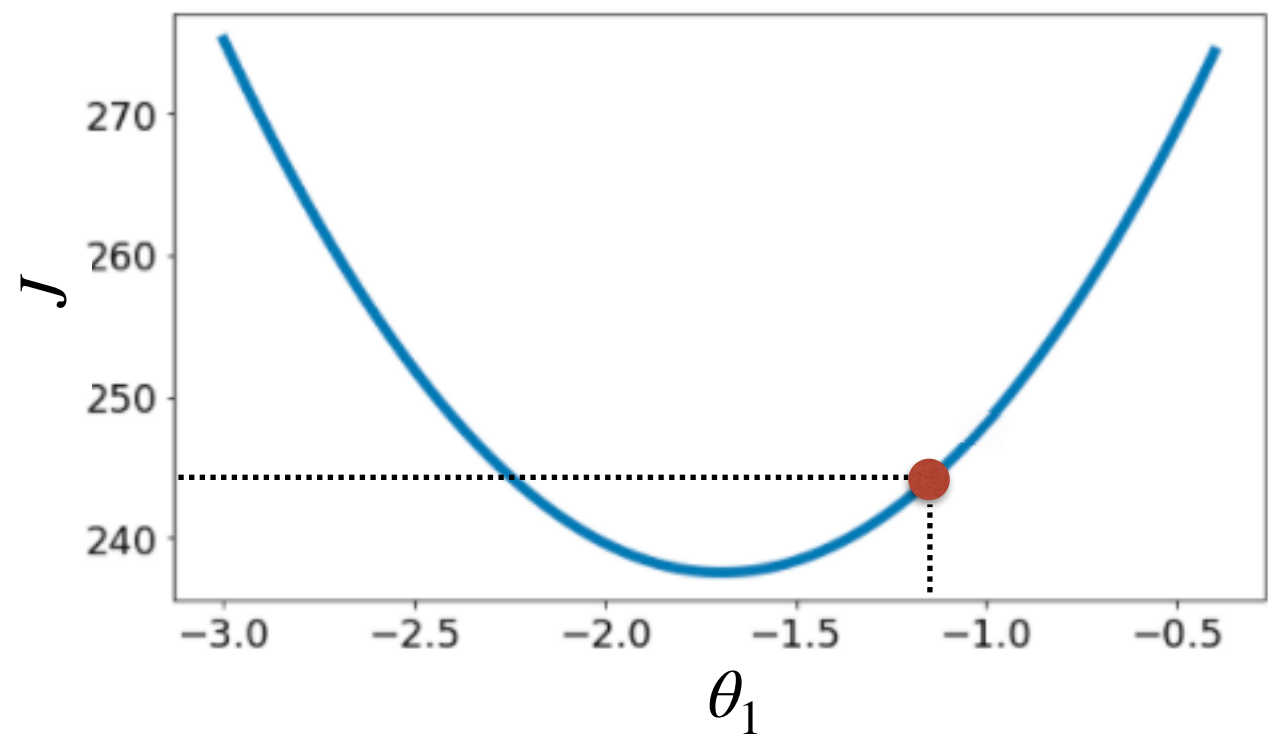
Regression Problems



Hypothesis:
 $f(x) = \theta_0 + \theta_1 x$

Assume $\theta_0 = 52.0$
 $\theta_1 = -1.0$

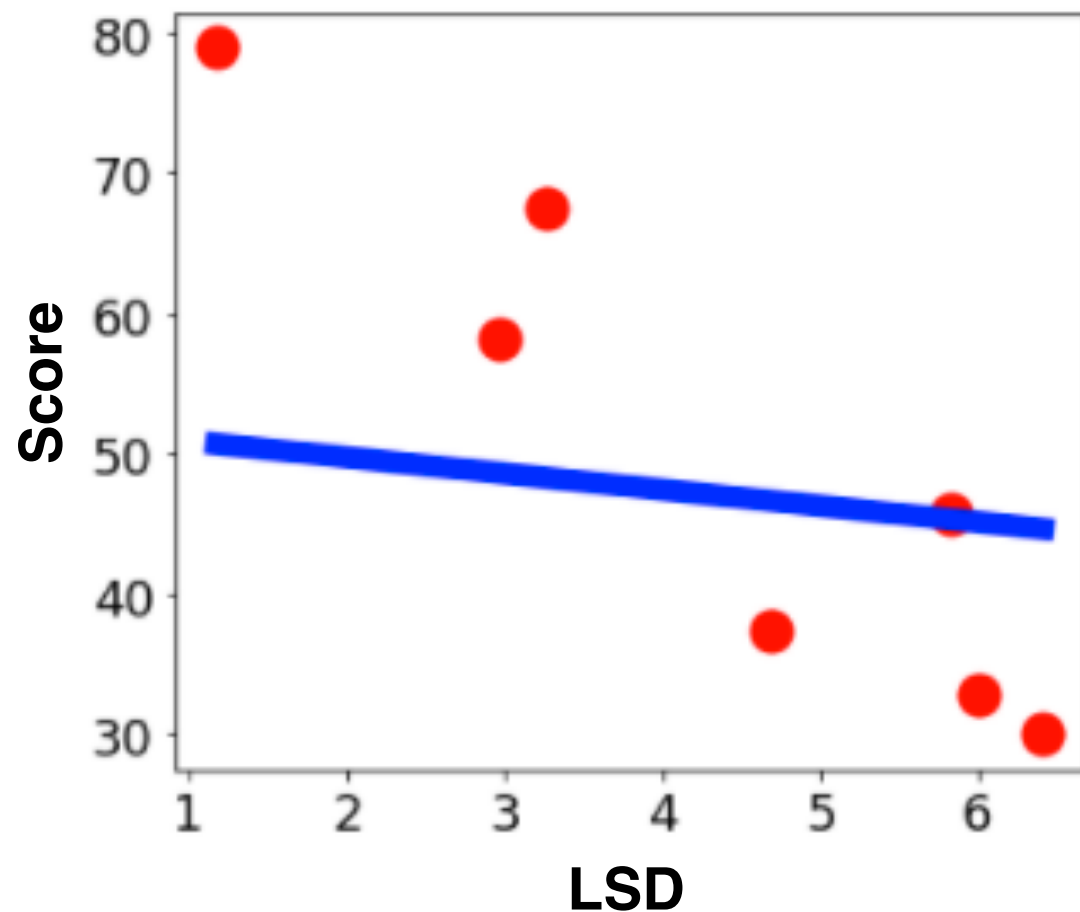
Let's keep $\theta_0 = 52.0$ fixed.
What happens with J as we change θ_1 ?



How should we update
 θ_1 to decrease the error J ?

Make θ_1 more negative

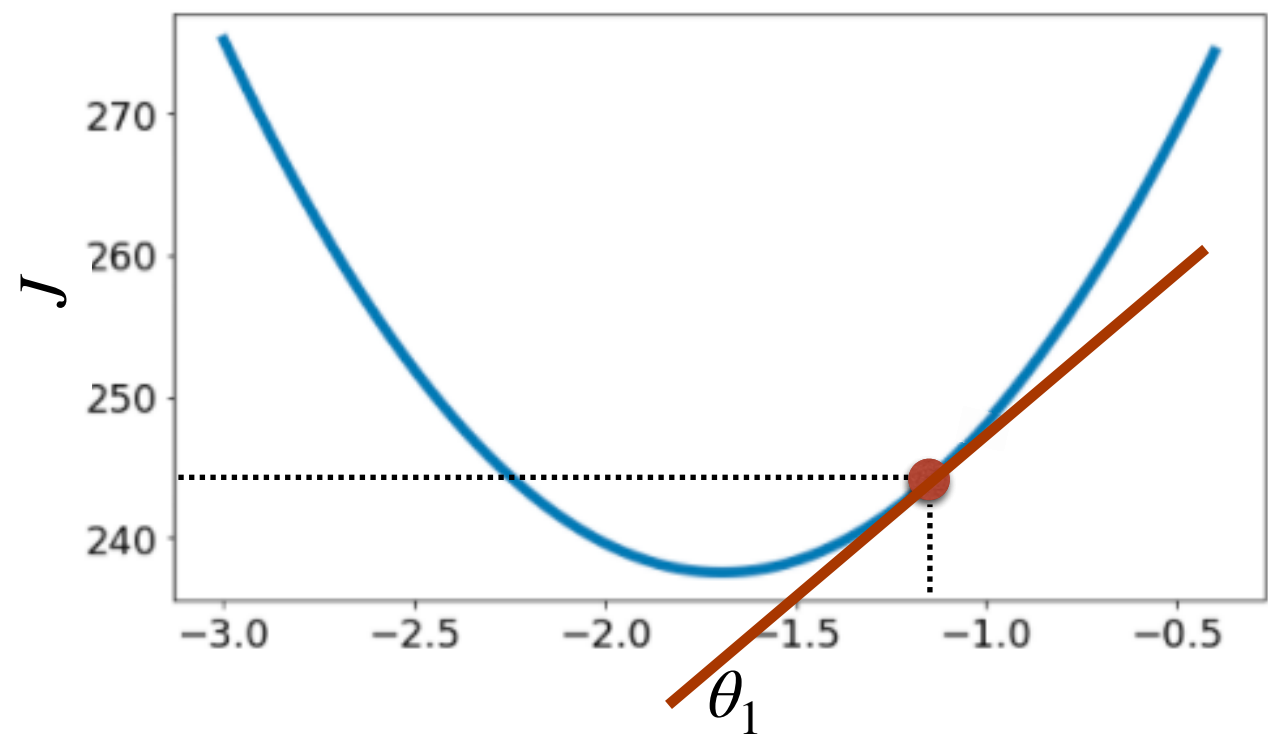
Regression Problems



Hypothesis:
 $f(x) = \theta_0 + \theta_1 x$

Assume $\theta_0 = 52.0$
 $\theta_1 = -1.0$

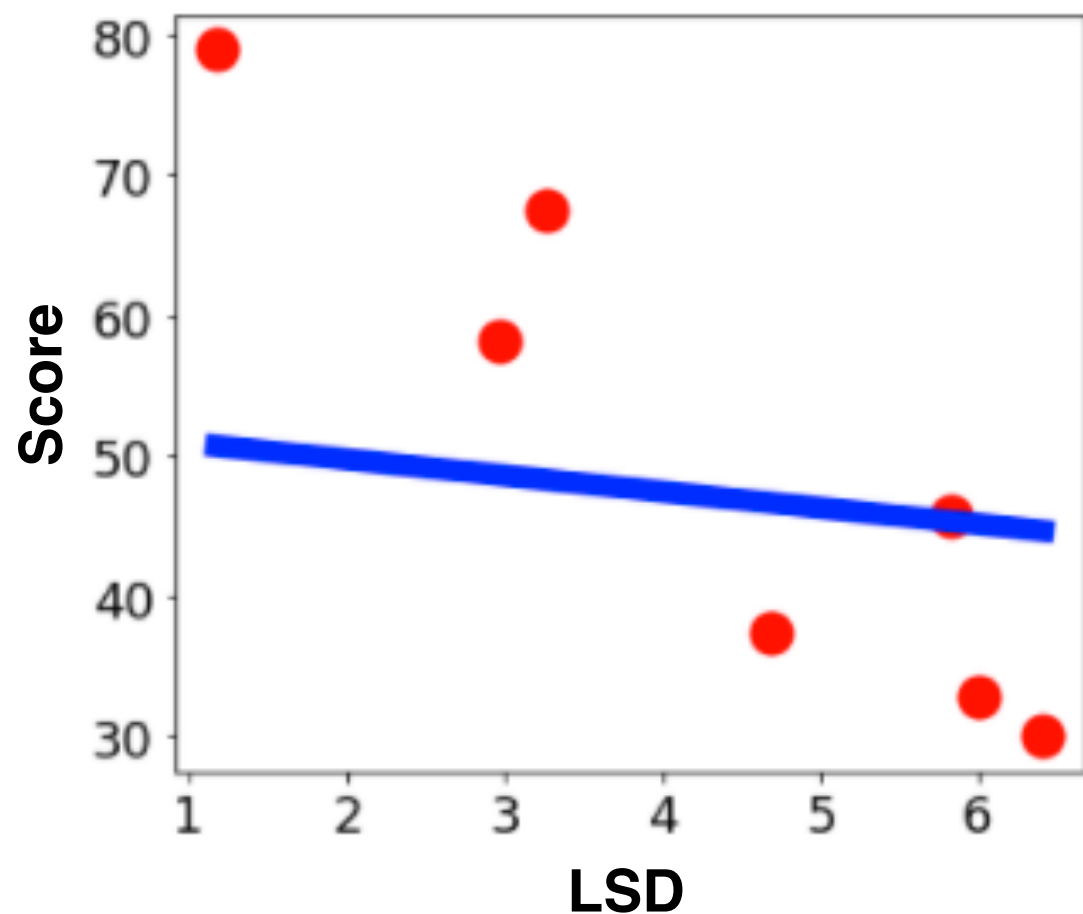
Let's keep $\theta_0 = 52.0$ fixed.
What happens with J as we change θ_1 ?



Make θ_1 more negative

Derivative of the J function with respect to θ_1
Slope of the function J at that point

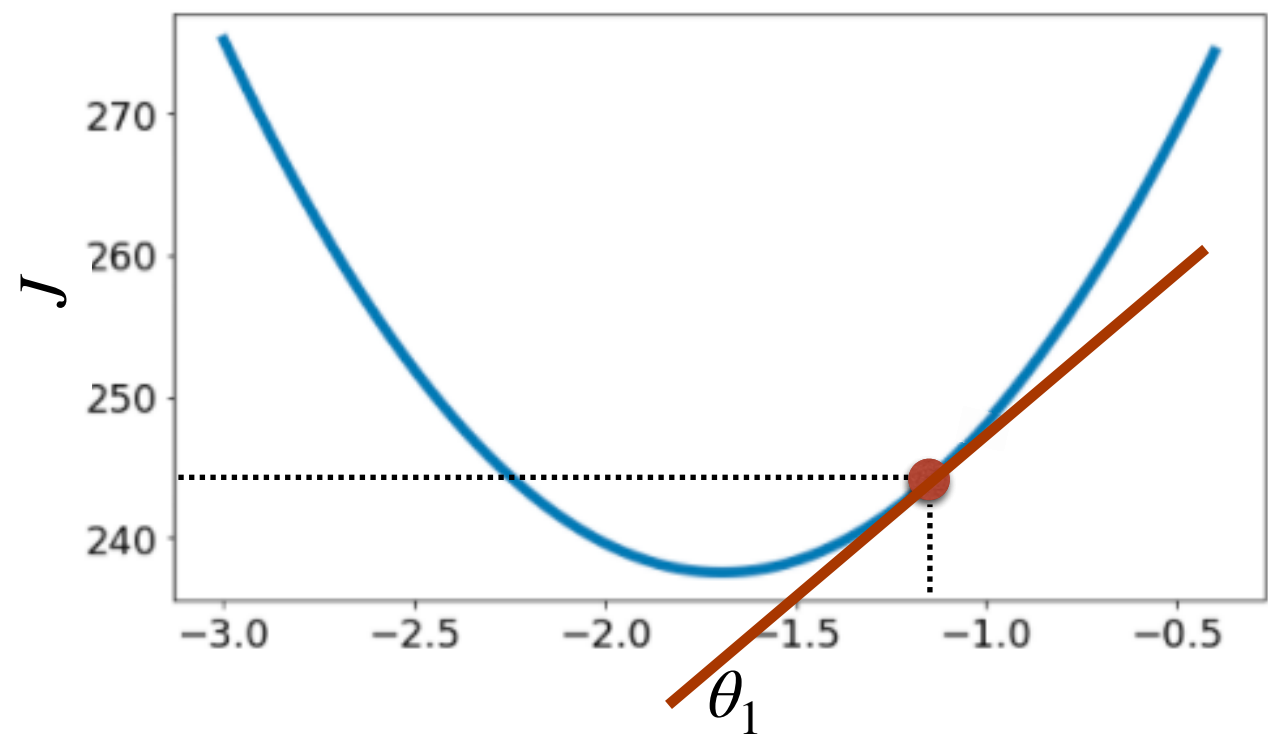
Regression Problems



Hypothesis:
 $f(x) = \theta_0 + \theta_1 x$

Assume $\theta_0 = 52.0$
 $\theta_1 = -1.0$

Let's keep $\theta_0 = 52.0$ fixed.
What happens with J as we change θ_1 ?



Make θ_1 more negative

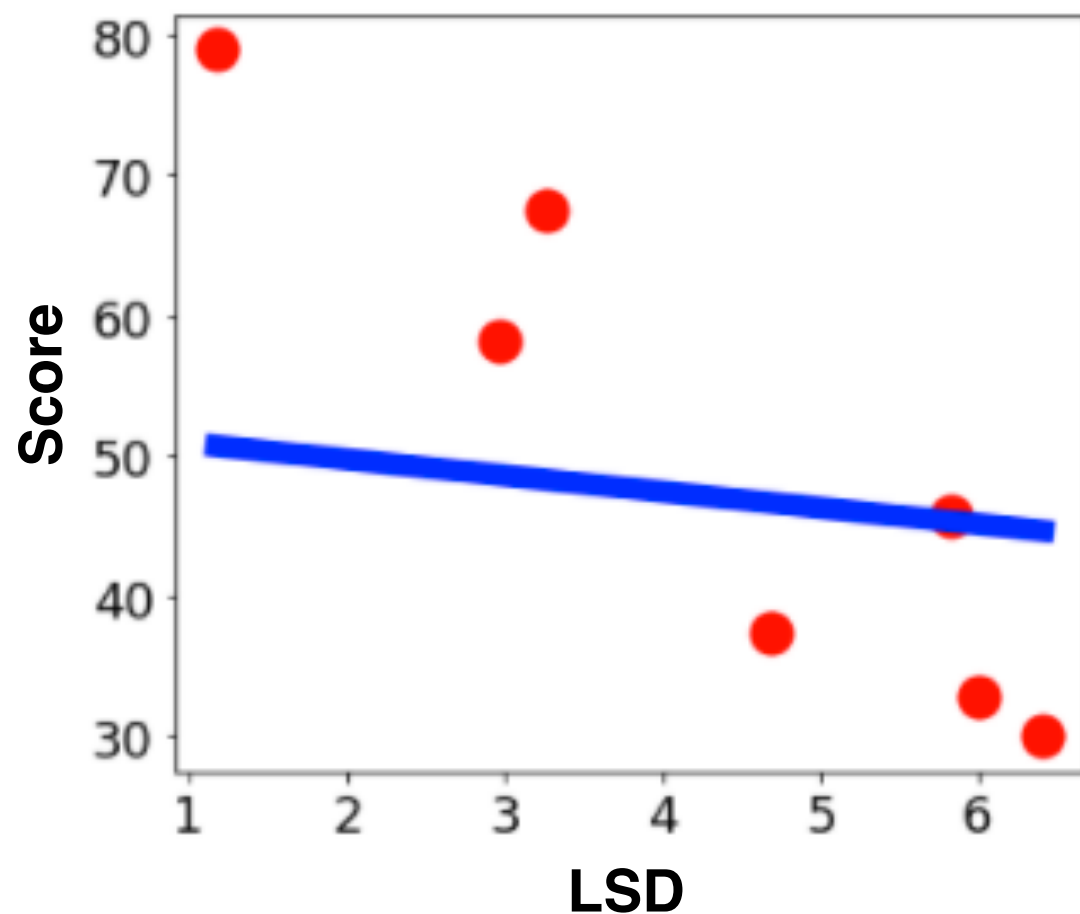
Derivative of the J function with respect to θ_1
 $= +2.42$ (positive derivative/slope)

→ if we make θ_1 more positive, J increases

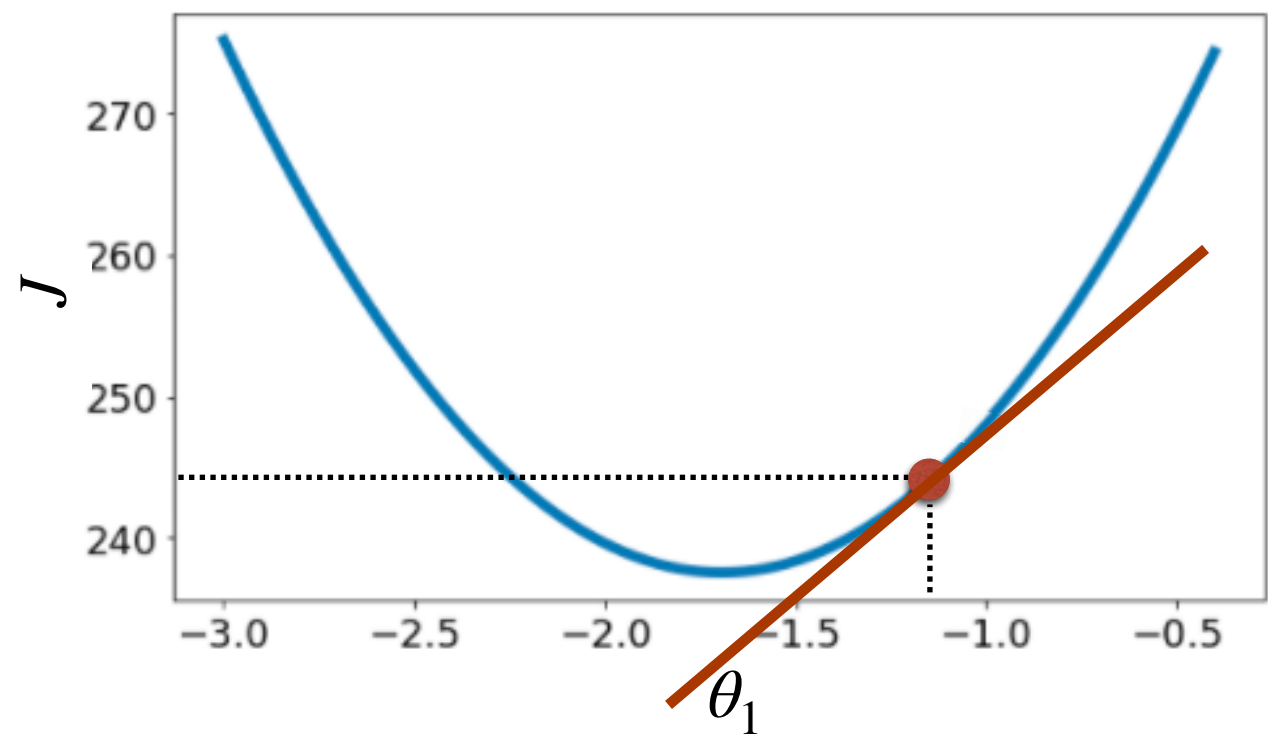
→ if we make θ_1 more negative, J decreases

→ if derivative is positive, we should make θ_1 more negative

Regression Problems



Let's keep $\theta_0 = 52.0$ fixed.
What happens with J as we change θ_1 ?

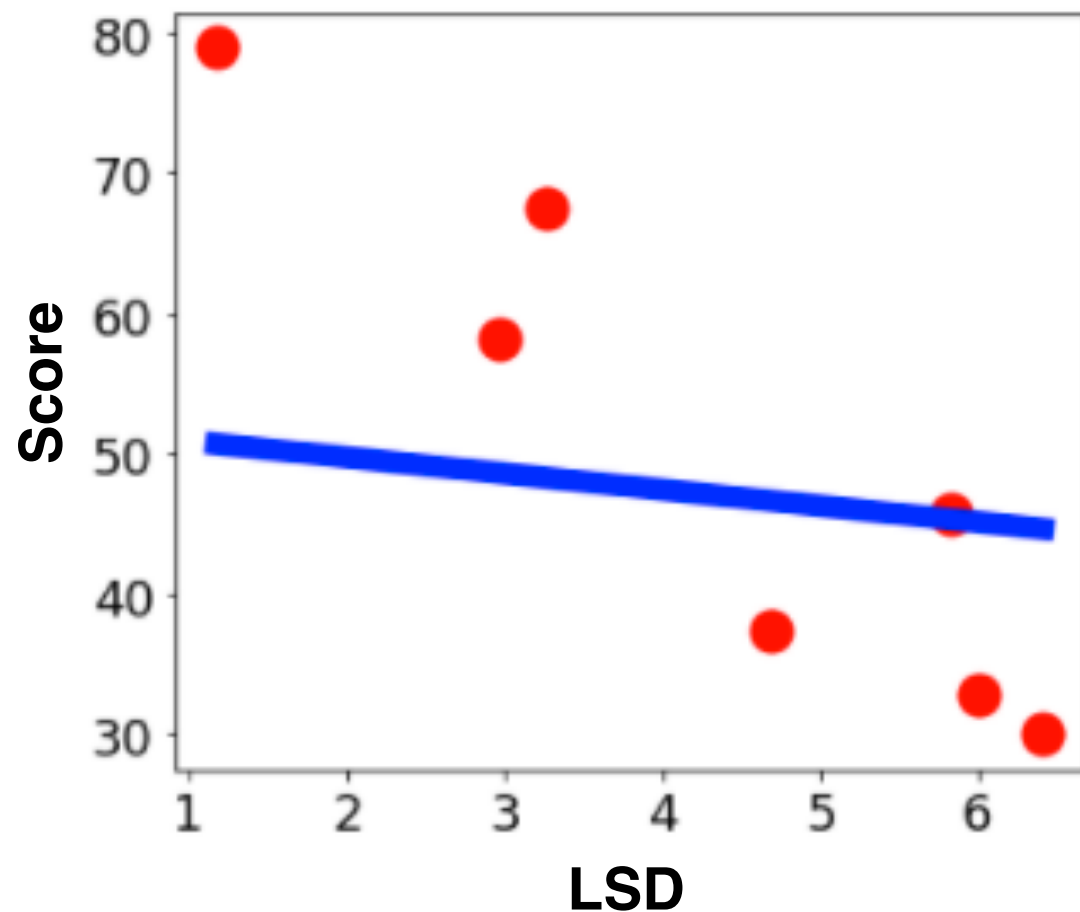


- if derivative is negative, we should make θ more positive
- if derivative is positive, we should make θ more negative

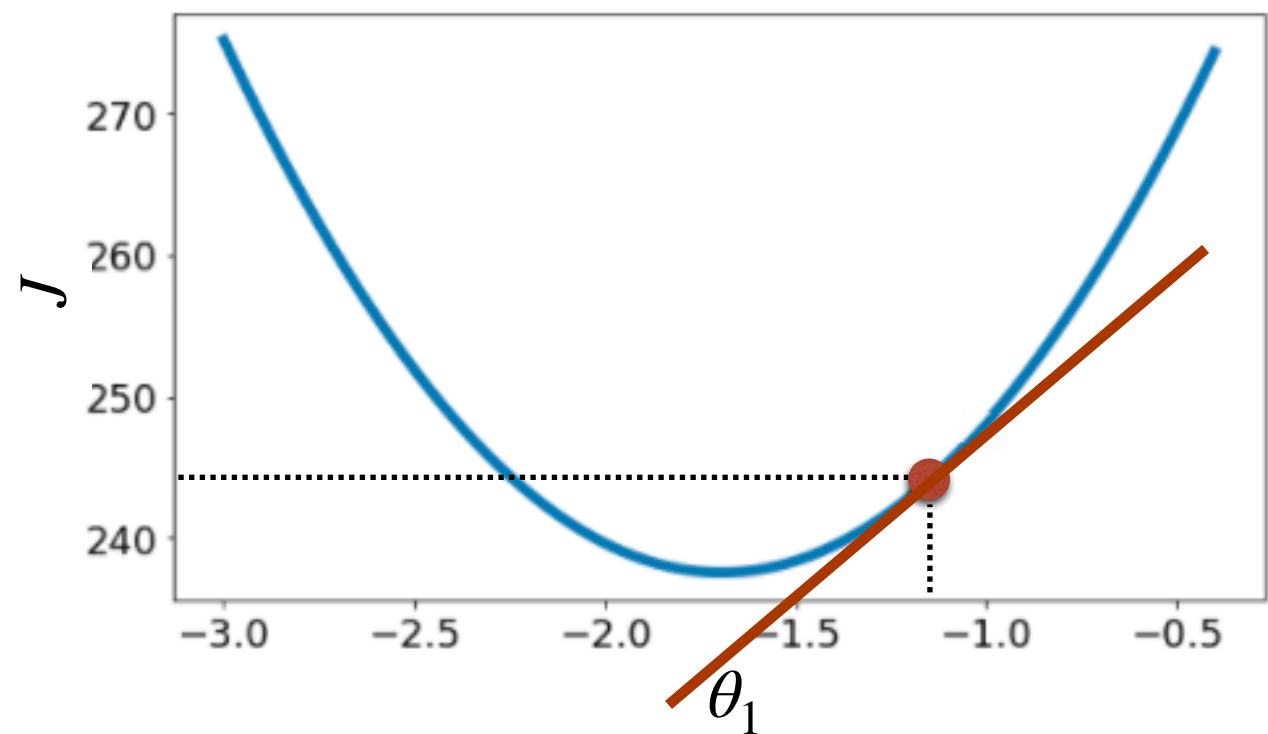
→ To decrease the error J , we should move each θ in the opposite direction of the derivative

Gradient Descent Algorithm

Regression Problems



Let's keep $\theta_0 = 52.0$ fixed.
What happens with J as we change θ_1 ?



→ To decrease the error J , we should move each θ in the opposite direction of the derivative

Gradient Descent Algorithm

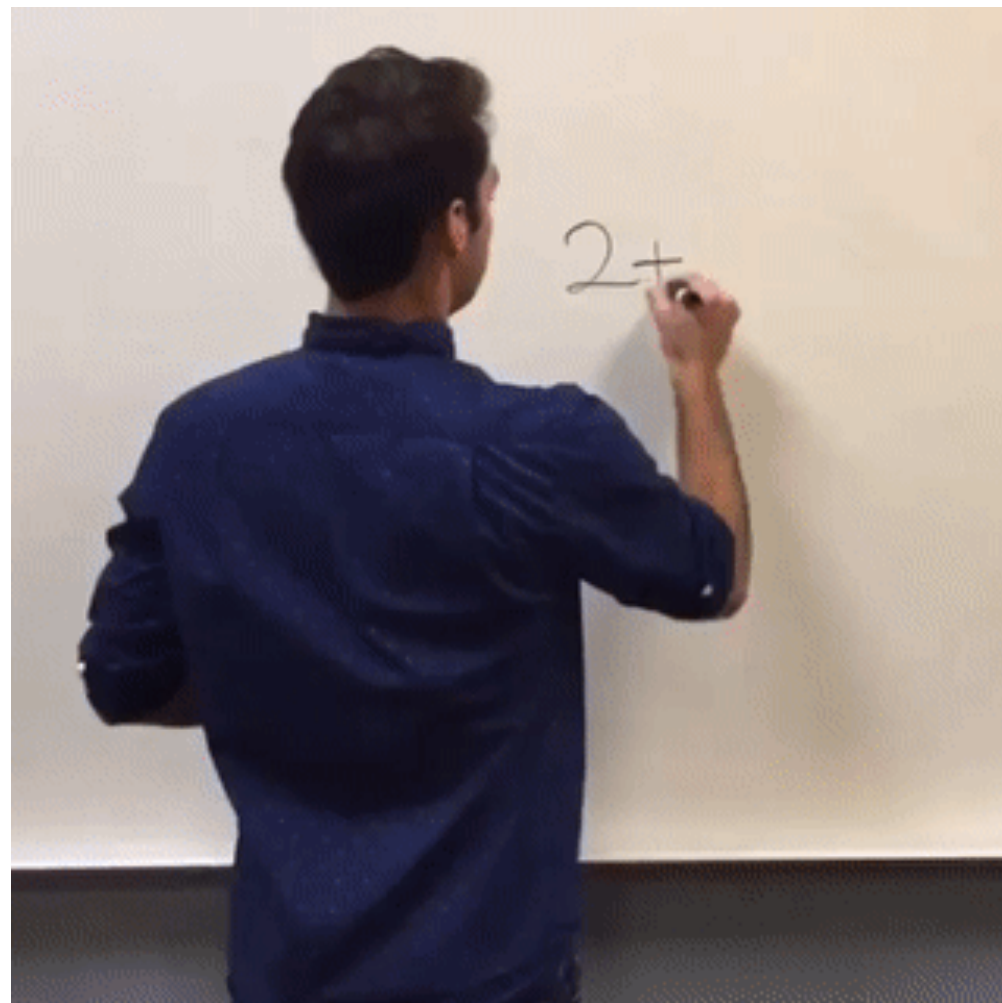
How to compute these derivatives, so that we know how to update each θ ?

Regression Problems

→ To decrease the error J , we should move each θ in the opposite direction of the derivative

Gradient Descent Algorithm

How to compute these derivatives, so that we know how to update each θ ?



Gradient Descent

We wish to find values for θ_0 and θ_1 that minimize the Mean Squared Error:

$$J(\theta_0, \theta_1) = \frac{1}{2n} \sum_{i=1}^n (f_{\theta}(x^{(i)}) - y^{(i)})^2 \quad \text{where } f_{\theta}(x) = \theta_0 + \theta_1 x \quad \text{is our hypothesis}$$

To decrease the error J , we should move each θ in the opposite direction of the derivative

$$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

$$\theta_0 := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

$$\frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$$

$$\theta_1 := \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$$

Gradient Descent

We wish to find values for θ_0 and θ_1 that minimize the **Mean Squared Error**:

$$J(\theta_0, \theta_1) = \frac{1}{2n} \sum_{i=1}^n (f_{\theta}(x^{(i)}) - y^{(i)})^2 \quad \text{where } f_{\theta}(x) = \theta_0 + \theta_1 x \quad \text{is our hypothesis}$$

To decrease the error **J**, we should move each θ in the opposite direction of the derivative

$$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) \qquad \theta_0 := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

Assume $\theta_0 = 52.0$
 $\theta_1 = -1.0$

Derivative of **J** with respect to θ_0
= -15.22 (negative derivative/slope)

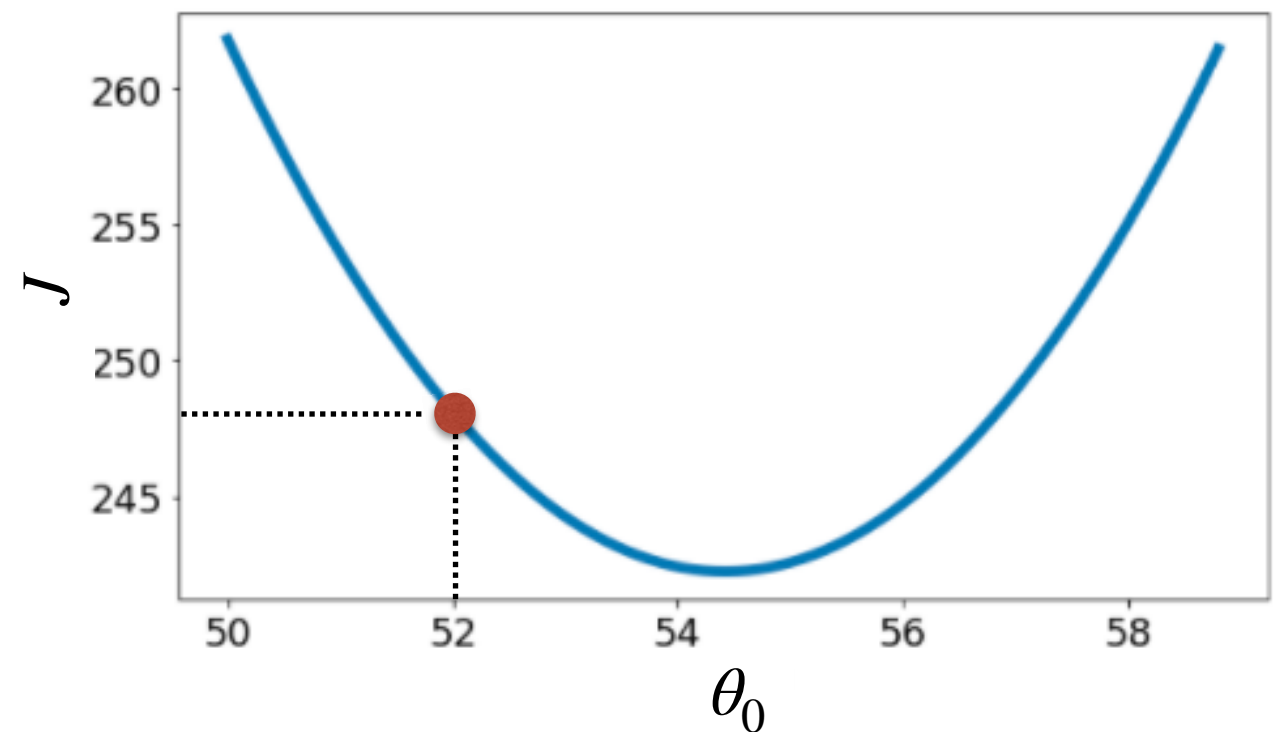
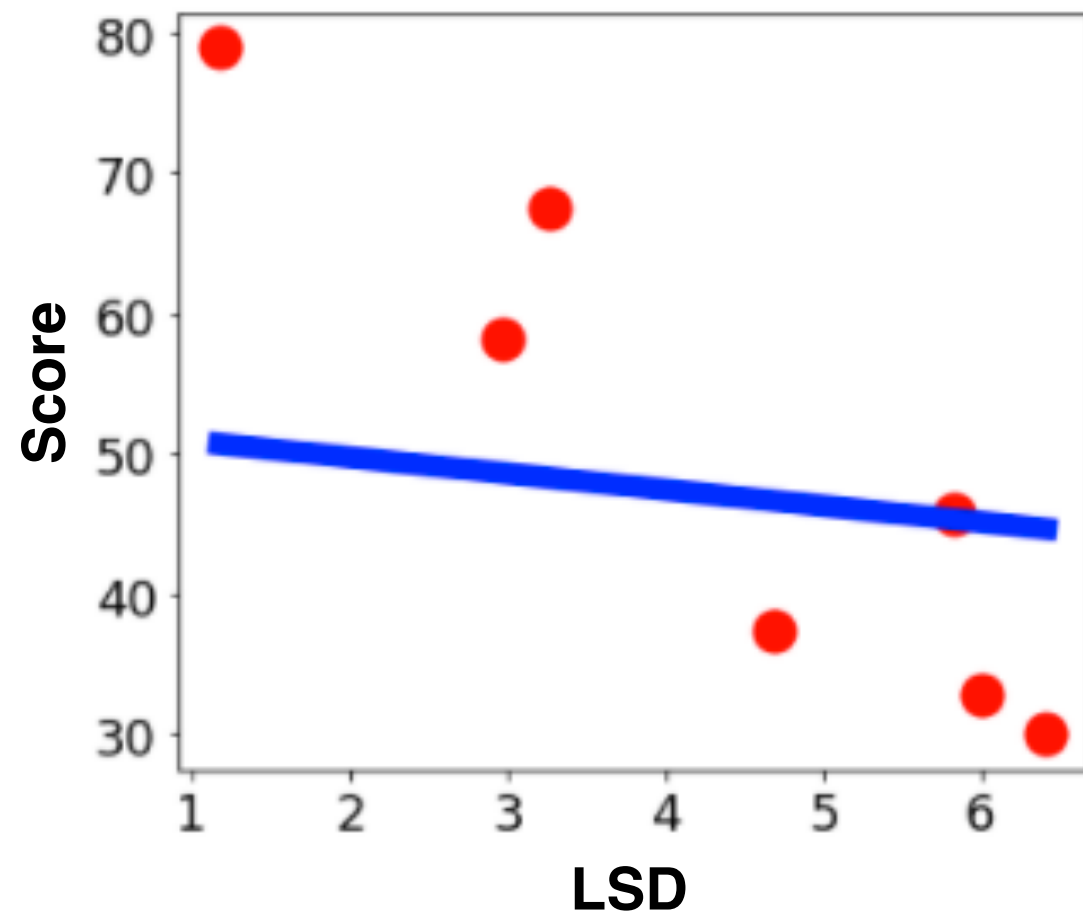

$$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = -15.22$$

→ if we make θ_0 more positive, **J** decreases

$$\begin{aligned} \theta_0 &:= 52.0 - \alpha(-15.22) \\ &:= 53.52 \end{aligned}$$

(assuming
 $\alpha=0.1$)

Selecting the step size α



Hypothesis:
 $f(x) = \theta_0 + \theta_1 x$

Current
parameters

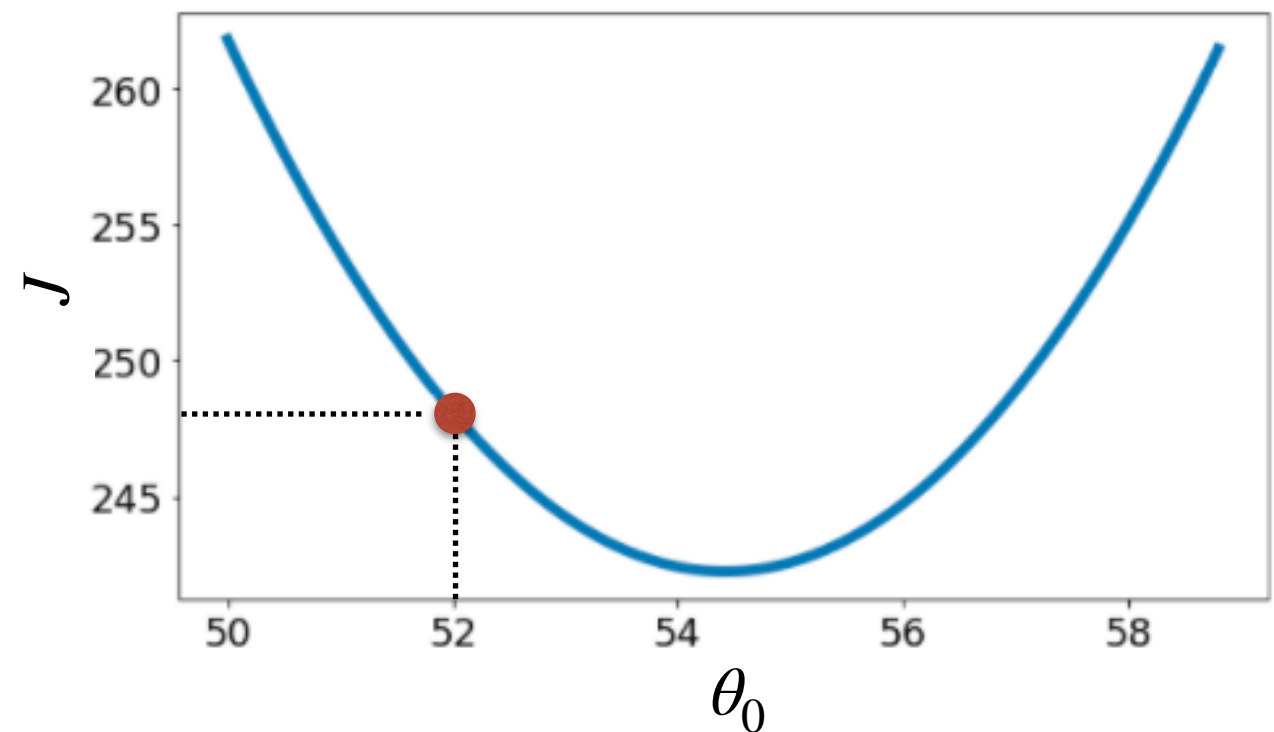
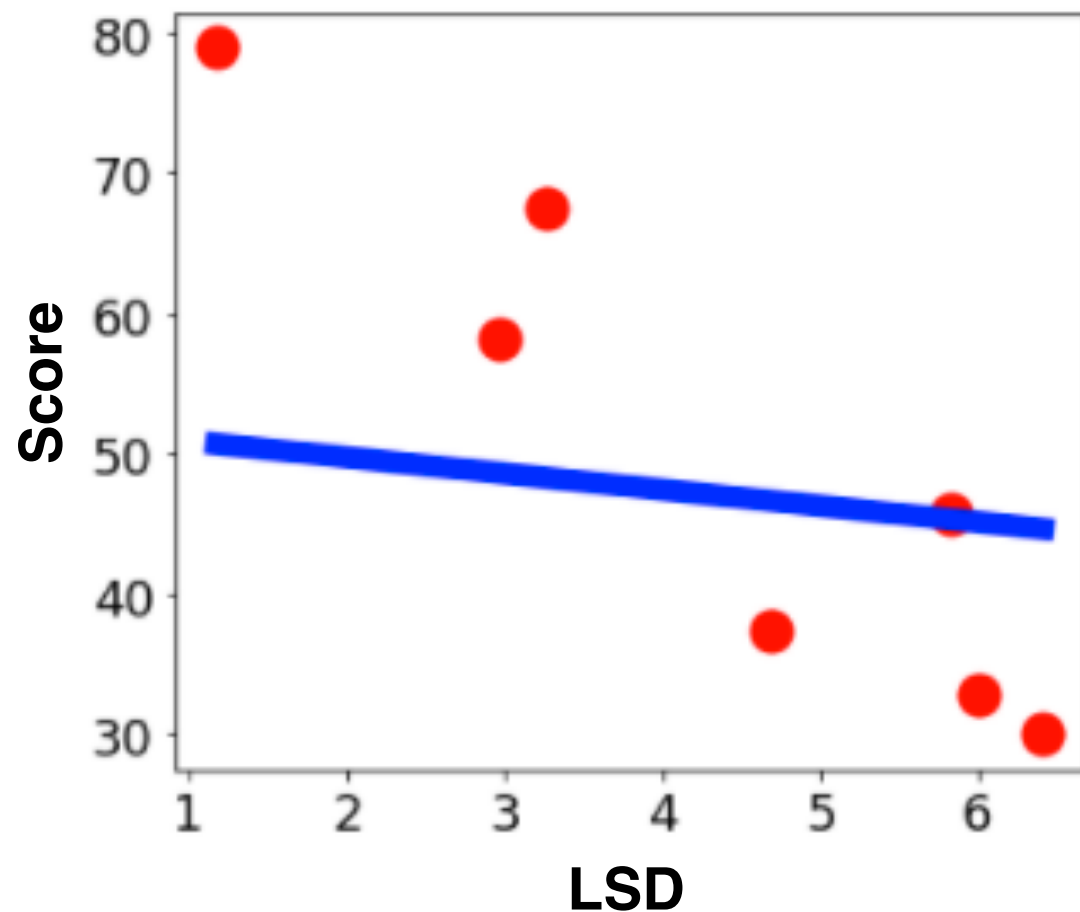
$\theta_0 = 52.0$
 $\theta_1 = -1.0$

$$\theta_0 := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

= -15.22

Which value of α should we use?

Selecting the step size α



Hypothesis:
 $f(x) = \theta_0 + \theta_1 x$

Current
parameters

$\theta_0 = 52.0$
 $\theta_1 = -1.0$

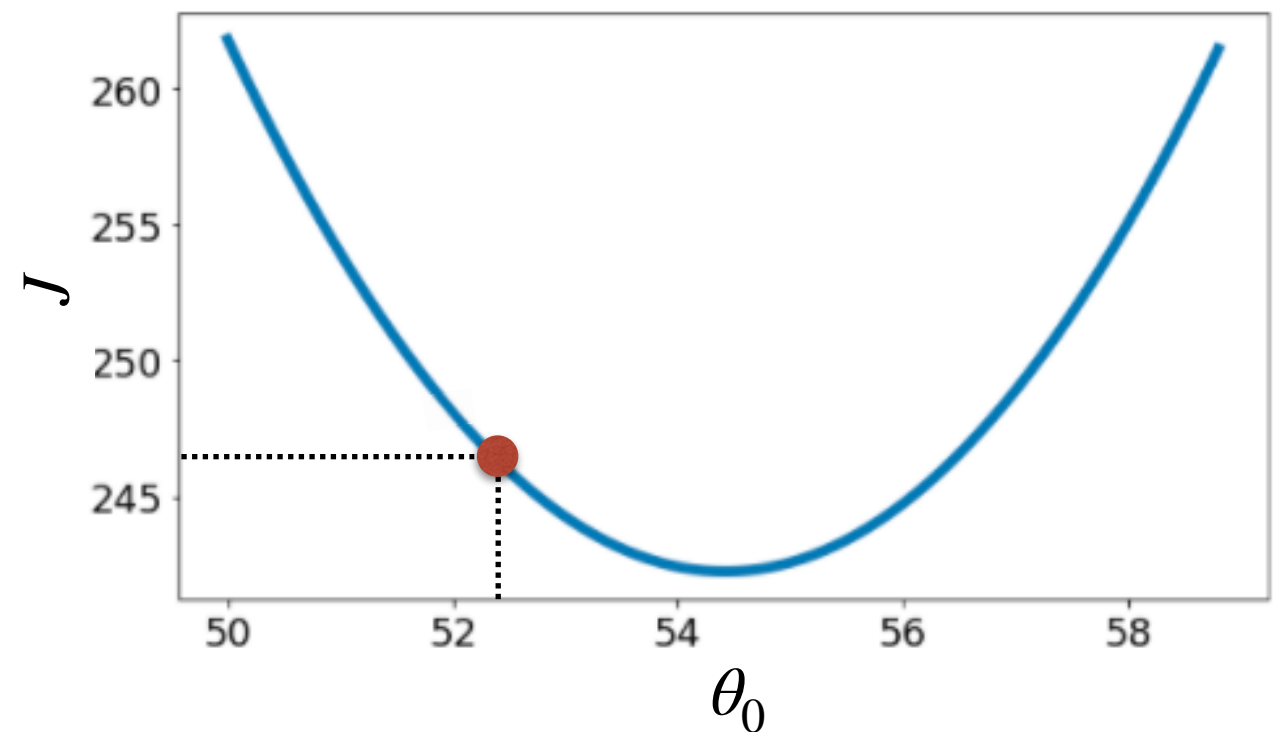
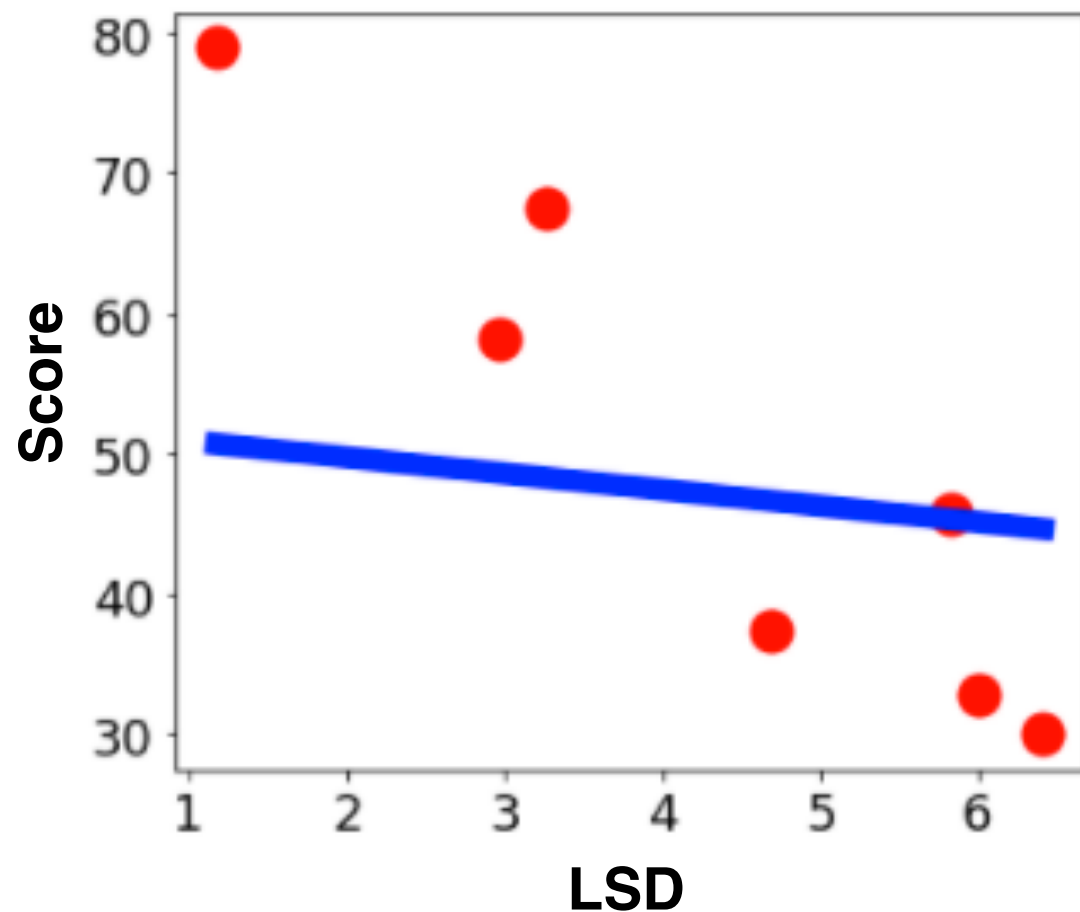
$$\theta_0 := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

= -15.22

$$\begin{aligned} \theta_0 &:= 52.0 - \alpha(-15.22) \\ &:= 53.52 \end{aligned}$$

$\alpha=0.1$

Selecting the step size α



Hypothesis:
 $f(x) = \theta_0 + \theta_1 x$

Current
parameters

$\theta_0 = 52.0$
 $\theta_1 = -1.0$

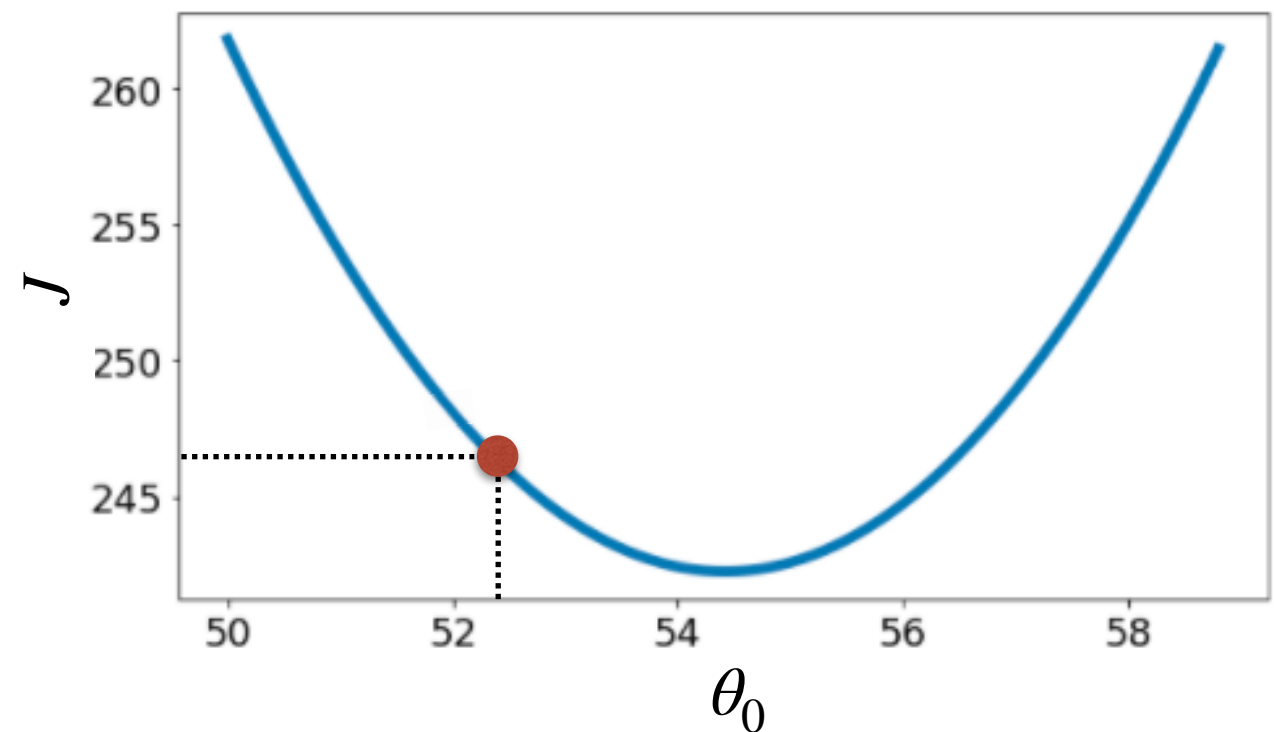
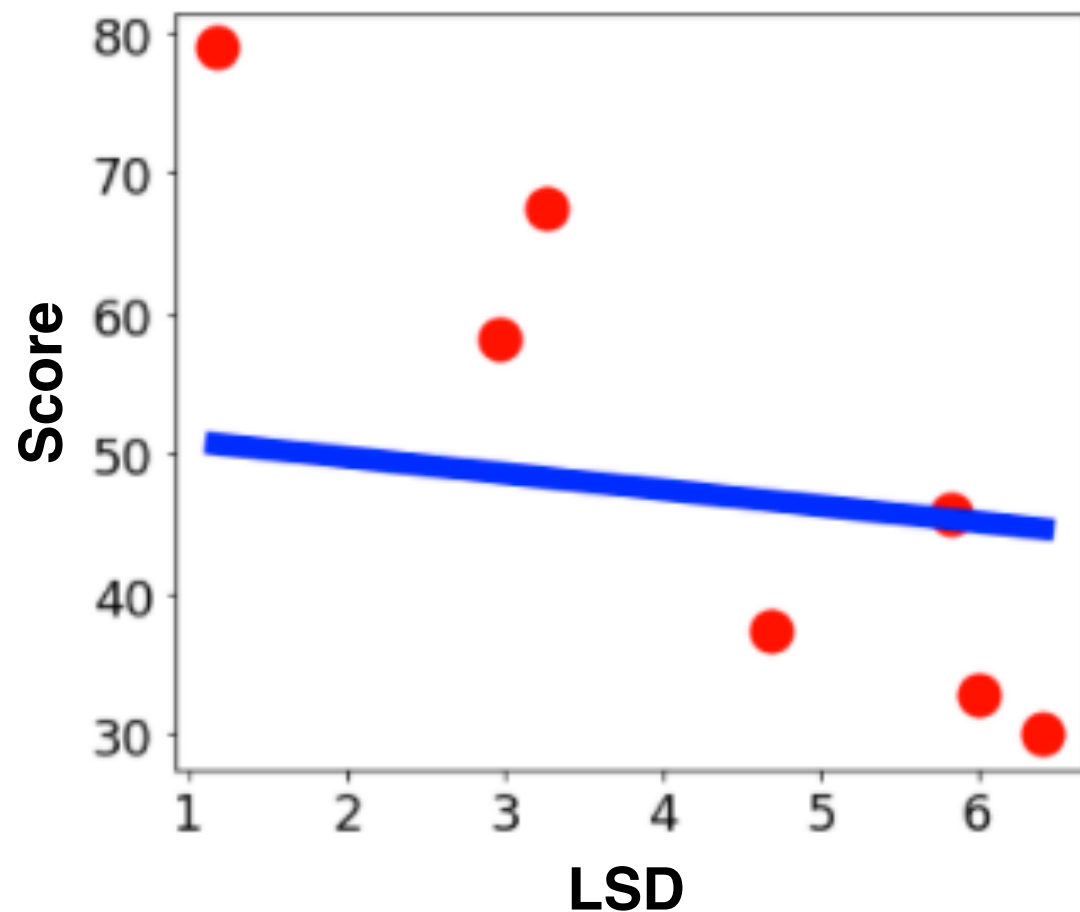
$$\theta_0 := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

= -15.22

$$\begin{aligned} \theta_0 &:= 52.0 - \alpha(-15.22) \\ &:= 53.52 \end{aligned}$$

$\alpha=0.1$

Selecting the step size α



Hypothesis:
 $f(x) = \theta_0 + \theta_1 x$

Current
parameters

$\theta_0 = 52.0$
 $\theta_1 = -1.0$

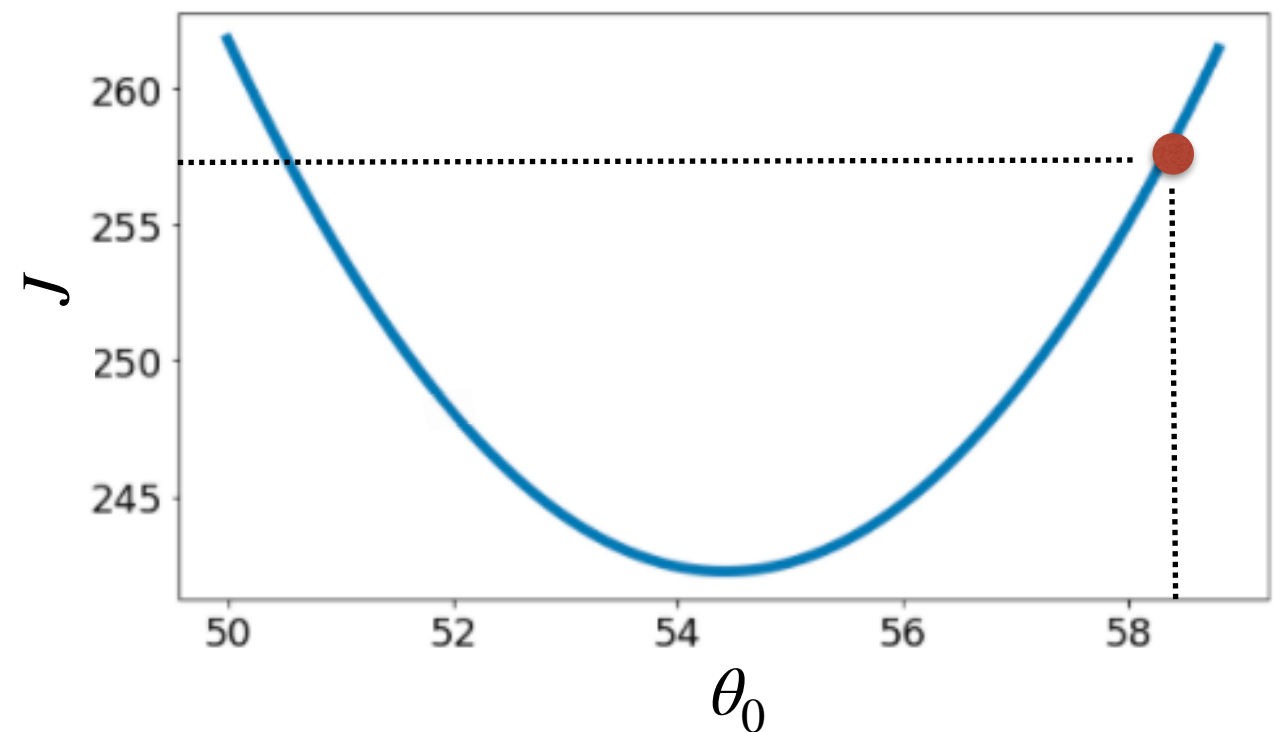
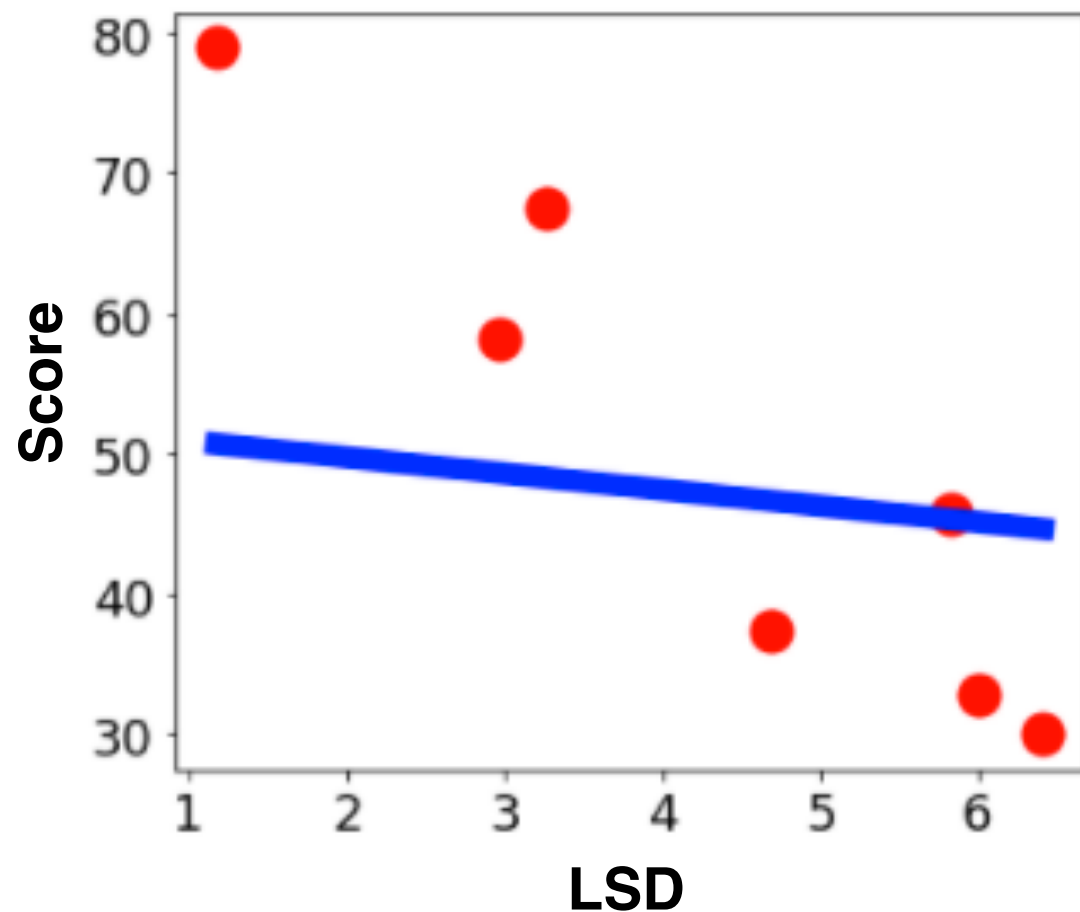
$$\theta_0 := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

= -15.22

$$\begin{aligned} \theta_0 &:= 52.0 - \alpha(-15.22) \\ &:= 59.61 \end{aligned}$$

$\alpha=0.5$?

Selecting the step size α



Hypothesis:
 $f(x) = \theta_0 + \theta_1 x$

Current
parameters

$\theta_0 = 52.0$
 $\theta_1 = -1.0$

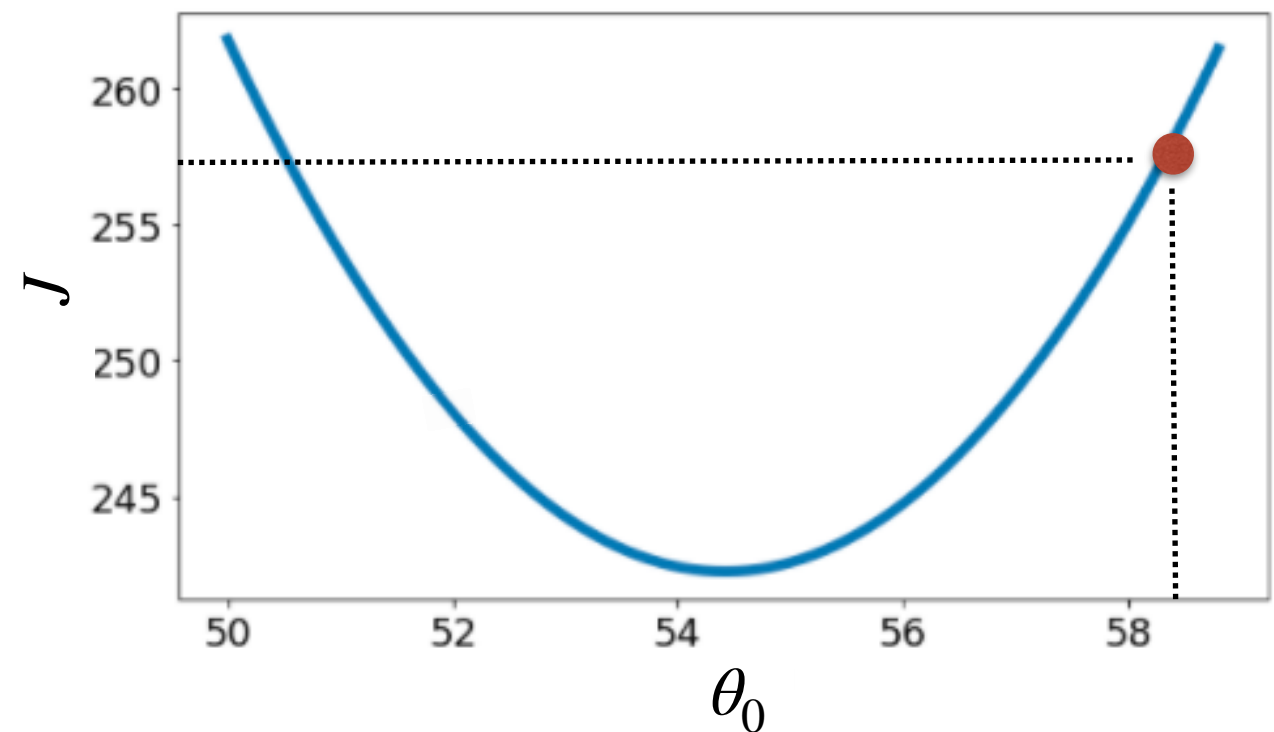
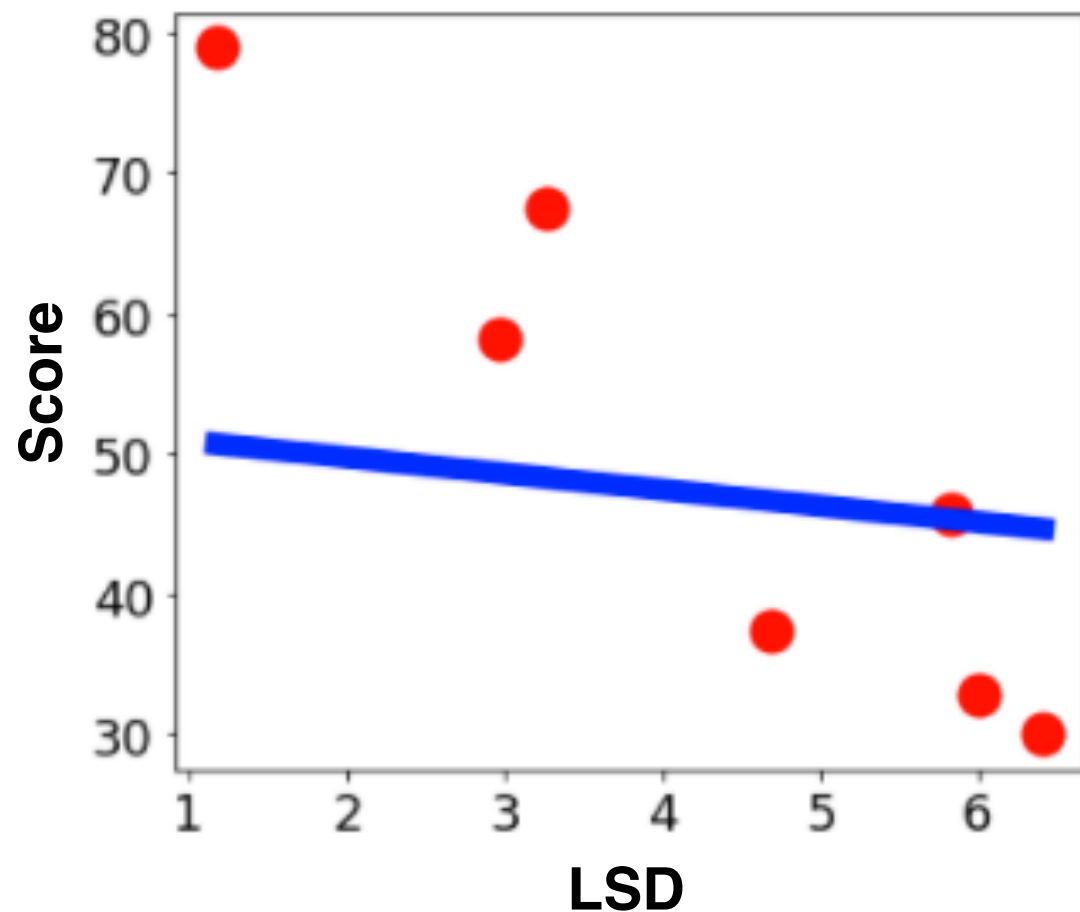
$$\theta_0 := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

= -15.22

$$\begin{aligned} \theta_0 &:= 52.0 - \alpha(-15.22) \\ &:= 59.61 \end{aligned}$$

$\alpha=0.5$?

Selecting the step size α



Hypothesis:
 $f(x) = \theta_0 + \theta_1 x$

Current
parameters

$\theta_0 = 52.0$
 $\theta_1 = -1.0$

$$\theta_0 := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

= -15.22

- overshooting

**step size α
needs to be sufficiently small**

Gradient Descent

We wish to find values for θ_0 and θ_1 that minimize the Mean Squared Error:

$$J(\theta_0, \theta_1) = \frac{1}{2n} \sum_{i=1}^n (f_{\theta}(x^{(i)}) - y^{(i)})^2 \quad \text{where } f_{\theta}(x) = \theta_0 + \theta_1 x \quad \text{is our hypothesis}$$

To decrease the error J , we should move each θ in the opposite direction of the derivative

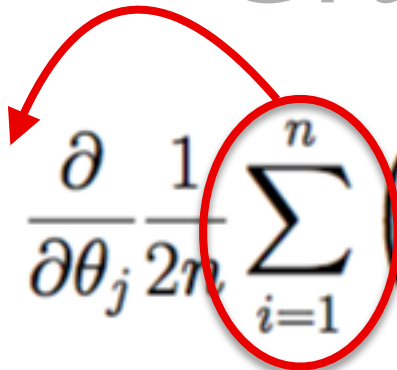
$$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = ?$$

$$\frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1) = ?$$

Gradient Descent

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) = \frac{\partial}{\partial \theta_j} \frac{1}{2n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right)^2$$

Gradient Descent

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) = \frac{\partial}{\partial \theta_j} \frac{1}{2n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right)^2$$


Gradient Descent

$$\begin{aligned}\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) &= \frac{\partial}{\partial \theta_j} \frac{1}{2n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right)^2 \\ &= \frac{1}{2n} \sum_{i=1}^n \frac{\partial}{\partial \theta_j} \left(f_{\theta}(x^{(i)}) - y^{(i)} \right)^2\end{aligned}$$

$$\frac{\partial}{\partial x} \left(\star \right)^2 = 2 \star \frac{\partial}{\partial x} \left(\star \right)$$

Chain Rule

Gradient Descent

$$\begin{aligned}\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) &= \frac{\partial}{\partial \theta_j} \frac{1}{2n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right)^2 \\ &= \frac{1}{2n} \sum_{i=1}^n \frac{\partial}{\partial \theta_j} \left(f_{\theta}(x^{(i)}) - y^{(i)} \right)^2 \\ &= \frac{1}{2n} \sum_{i=1}^n 2 \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(f_{\theta}(x^{(i)}) - y^{(i)} \right)\end{aligned}$$

$$\frac{\partial}{\partial x} (\star)^2 = 2 \star \frac{\partial}{\partial x} (\star)$$

Chain Rule

Gradient Descent

$$\begin{aligned}\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) &= \frac{\partial}{\partial \theta_j} \frac{1}{2n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right)^2 \\ &= \frac{1}{2n} \sum_{i=1}^n \frac{\partial}{\partial \theta_j} \left(f_{\theta}(x^{(i)}) - y^{(i)} \right)^2 \\ &= \frac{1}{\cancel{2n}} \sum_{i=1}^n \cancel{2} \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(f_{\theta}(x^{(i)}) - y^{(i)} \right)\end{aligned}$$

Gradient Descent

$$\begin{aligned}\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) &= \frac{\partial}{\partial \theta_j} \frac{1}{2n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right)^2 \\&= \frac{1}{2n} \sum_{i=1}^n \frac{\partial}{\partial \theta_j} \left(f_{\theta}(x^{(i)}) - y^{(i)} \right)^2 \\&= \frac{1}{\cancel{2}n} \sum_{i=1}^n \cancel{2} \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \\&= \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\textcircled{f_{\theta}(x^{(i)})} - y^{(i)} \right)\end{aligned}$$

Gradient Descent

$$\begin{aligned}\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) &= \frac{\partial}{\partial \theta_j} \frac{1}{2n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right)^2 \\&= \frac{1}{2n} \sum_{i=1}^n \frac{\partial}{\partial \theta_j} \left(f_{\theta}(x^{(i)}) - y^{(i)} \right)^2 \\&= \frac{1}{\cancel{2n}} \sum_{i=1}^n \cancel{2} \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \\&= \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \\&= \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\theta_0 + \theta_1 x^{(i)} - y^{(i)} \right)\end{aligned}$$

Gradient Descent

$$\begin{aligned}\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) &= \frac{\partial}{\partial \theta_j} \frac{1}{2n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right)^2 \\&= \frac{1}{2n} \sum_{i=1}^n \frac{\partial}{\partial \theta_j} \left(f_{\theta}(x^{(i)}) - y^{(i)} \right)^2 \\&= \frac{1}{\cancel{2}n} \sum_{i=1}^n \cancel{2} \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \\&= \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \\&= \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\theta_0 + \theta_1 x^{(i)} - \cancel{y^{(i)}} \right)\end{aligned}$$

Gradient Descent

$$\begin{aligned}\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) &= \frac{\partial}{\partial \theta_j} \frac{1}{2n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right)^2 \\&= \frac{1}{2n} \sum_{i=1}^n \frac{\partial}{\partial \theta_j} \left(f_{\theta}(x^{(i)}) - y^{(i)} \right)^2 \\&= \cancel{\frac{1}{2n}} \sum_{i=1}^n \cancel{2} \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \\&= \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \\&= \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\theta_0 + \theta_1 x^{(i)} - y^{(i)} \right) \\&= \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\theta_0 + \theta_1 x^{(i)} \right)\end{aligned}$$

Gradient Descent

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\theta_0 + \theta_1 x^{(i)} \right)$$

$$j = 0 : \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \text{ x ?}$$

$$j = 1 : \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \text{ x ?}$$

Gradient Descent

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\theta_0 + \theta_1 x^{(i)} \right)$$

$j = 0$: $\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) 1$

$j = 1$: $\frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) x \text{ ?}$

Gradient Descent

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\theta_0 + \theta_1 x^{(i)} \right)$$

$$j = 0 : \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) 1$$

$$j = 1 : \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) x^{(i)}$$

Gradient Descent

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\theta_0 + \theta_1 x^{(i)} \right)$$

$$j = 0 \quad : \quad \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) 1$$

$$j = 1 \quad : \quad \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) x^{(i)}$$

$$\theta_0 := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

$$\theta_1 := \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$$

Gradient Descent

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\theta_0 + \theta_1 x^{(i)} \right)$$

$$j = 0 : \quad \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) 1$$

$$j = 1 : \quad \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) x^{(i)}$$

$$\theta_0 := \theta_0 - \alpha \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) 1$$

$$\theta_1 := \theta_1 - \alpha \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) x^{(i)}$$

Implementation

Hypothesis:

$$f_{\theta}(x) = \theta_0 + \theta_1 x$$

**Error function
to be minimized:**

$$J(\theta_0, \theta_1) = \frac{1}{2n} \sum_{i=1}^n (f_{\theta}(x^{(i)}) - y^{(i)})^2$$

**Gradient descent
weight update rule**

$$\theta_0 := \theta_0 - \alpha \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) 1$$

$$\theta_1 := \theta_1 - \alpha \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) x^{(i)}$$

Regression Problems

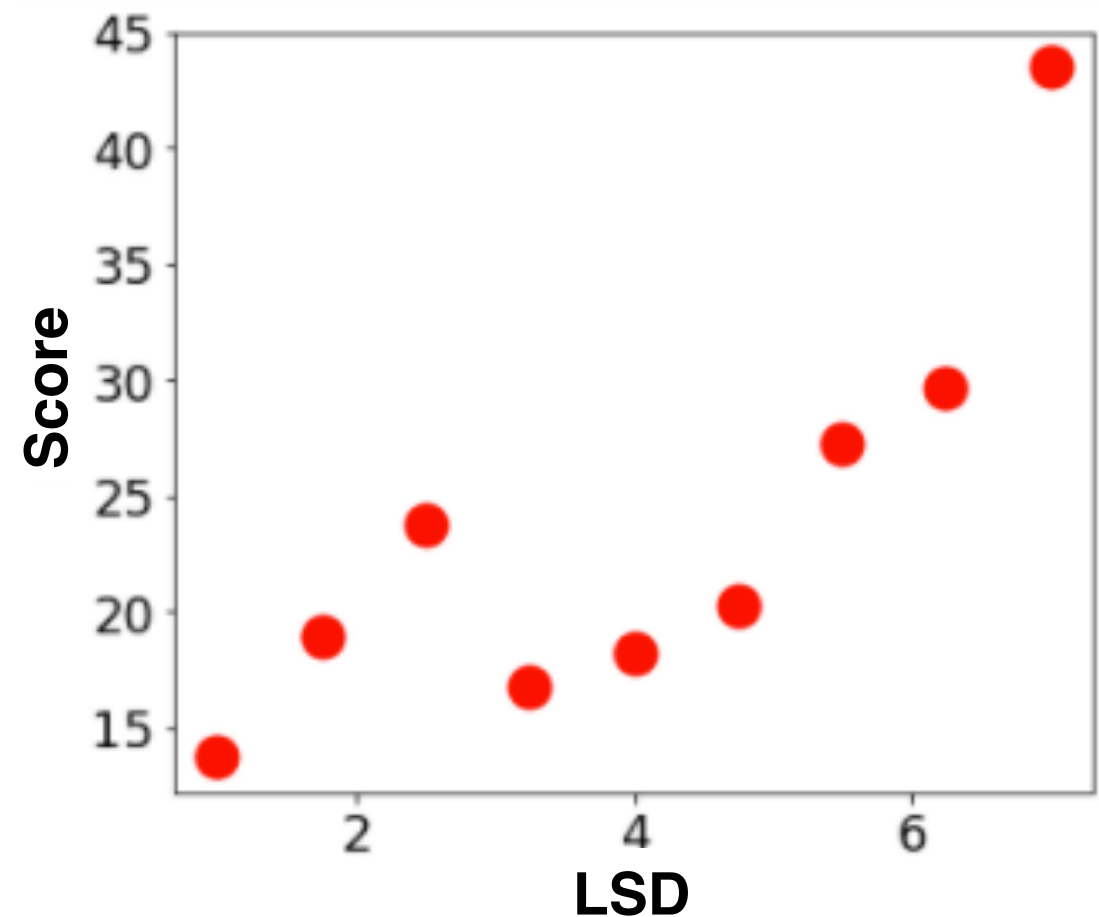
What if the data cannot be approximated well by a straight line?

	x =LSD Dosage	y = Exam Score
Student1	1.0	13.69
Student2	1.75	18.98
Student3	2.5	23.79
Student4	3.25	16.80
Student5	4.0	18.15
Student6	4.75	20.22
Student7	5.5	27.25
Student8	6.25	29.64
Student9	7.0	43.47

$f(\text{LSD_Dosage}) \rightarrow \text{Exam_Score}$

$f(x) \rightarrow y$

We can make a more complex hypothesis



Hypothesis:

$$f(x) = \theta_0 + \theta_1 x + \theta_2 \sin(x)$$

Gradient Descent

We wish to find values for θ_0 and θ_1 and θ_2 that minimize the Mean Squared Error:

$$J(\theta_0, \theta_1, \theta_2) = \frac{1}{2n} \sum_{i=1}^n (f_{\theta}(x^{(i)}) - y^{(i)})^2$$

where $f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 \sin(x)$ is our hypothesis

To decrease the error J , we should move each θ in the opposite direction of the derivative

$$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1, \theta_2) \qquad \theta_0 := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1, \theta_2)$$

$$\frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1, \theta_2) \qquad \theta_1 := \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1, \theta_2)$$

$$\frac{\partial}{\partial \theta_2} J(\theta_0, \theta_1, \theta_2) \qquad \theta_2 := \theta_2 - \alpha \frac{\partial}{\partial \theta_2} J(\theta_0, \theta_1, \theta_2)$$

Gradient Descent

We wish to find values for θ_0 and θ_1 and θ_2 that minimize the Mean Squared Error:

$$J(\theta_0, \theta_1, \theta_2) = \frac{1}{2n} \sum_{i=1}^n (f_{\theta}(x^{(i)}) - y^{(i)})^2$$

where $f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 \sin(x)$ is our hypothesis

To decrease the error J , we should move each θ in the opposite direction of the derivative

$$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1, \theta_2) = ?$$

$$\frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1, \theta_2) = ?$$

$$\frac{\partial}{\partial \theta_2} J(\theta_0, \theta_1, \theta_2) = ?$$

Hypothesis: $f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 \sin(x)$

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\theta_0 + \theta_1 x^{(i)} \right)$$



This was the case when the hypothesis was $f_{\theta}(x) = \theta_0 + \theta_1 x$

Hypothesis: $f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 \sin(x)$

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\theta_0 + \theta_1 x^{(i)} \right)$$



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Hypothesis: $f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 \sin(x)$

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\theta_0 + \theta_1 x^{(i)} + \theta_2 \sin(x^{(i)}) \right)$$



This is the case when the hypothesis is $f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 \sin(x)$

Hypothesis: $f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 \sin(x)$

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\theta_0 + \theta_1 x^{(i)} + \theta_2 \sin(x^{(i)}) \right)$$

$$j = 0 : \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) 1$$
$$j = 1 : \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) x^{(i)}$$

} like before

$$j = 2 : \frac{\partial}{\partial \theta_2} J(\theta_0, \theta_1, \theta_2) = ?$$

Hypothesis: $f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 \sin(x)$

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\theta_0 + \theta_1 x^{(i)} + \theta_2 \sin(x^{(i)}) \right)$$

$j = 0$: $\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) 1$

$j = 1$: $\frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) x^{(i)}$

$j = 2$: $\frac{\partial}{\partial \theta_2} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \times ?$

like before

Hypothesis: $f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 \sin(x)$

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\theta_0 + \theta_1 x^{(i)} + \theta_2 \sin(x^{(i)}) \right)$$

$$\begin{aligned} j = 0 & : \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) 1 \\ j = 1 & : \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) x^{(i)} \\ j = 2 & : \frac{\partial}{\partial \theta_2} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \sin(x^{(i)}) \end{aligned} \quad \left. \vphantom{\begin{aligned} j = 0 \\ j = 1 \\ j = 2 \end{aligned}} \right\} \text{like before}$$

Hypothesis: $f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 \sin(x)$

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What is the pattern in the update rules?

Hypothesis: $f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 \sin(x)$

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\theta_0 + \theta_1 x^{(i)} + \theta_2 \sin(x^{(i)}) \right)$$

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$$j = 2 : \frac{\partial}{\partial \theta_2} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \sin(x^{(i)})$$

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What is the pattern in the update rules?

Implementation

Hypothesis:

$$f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 \sin(x)$$

**Error function
to be minimized:**

$$J(\theta_0, \theta_1, \theta_2) = \frac{1}{2n} \sum_{i=1}^n (f_{\theta}(x^{(i)}) - y^{(i)})^2$$

**Gradient descent
weight update rule**

$$\theta_0 := \theta_0 - \alpha \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) 1$$

$$\theta_1 := \theta_1 - \alpha \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) x^{(i)}$$

$$\theta_2 := \theta_2 - \alpha \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \sin(x^{(i)})$$

What if the hypothesis was $f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 \sin(x)$?

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\theta_0 + \theta_1 x^{(i)} + \theta_2 \sin(x^{(i)}) \right)$$

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$$j = 1 : \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) x^{(i)}$$

$$j = 2 : \frac{\partial}{\partial \theta_2} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \sin(x^{(i)})$$

What is the pattern in the update rules?

What if the hypothesis was $f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2(x)^9$?

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\text{?} \right)$$

$$j = 0 : \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \mathbf{x} \text{ ?}$$

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$$j = 2 : \frac{\partial}{\partial \theta_2} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \mathbf{x} ?$$

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$$j = 2 : \frac{\partial}{\partial \theta_2} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \mathbf{x} ?$$

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$$j = 2 : \frac{\partial}{\partial \theta_2} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) (x^{(i)})^9$$

What is the pattern in the update rules?

Implementation

Hypothesis:

$$f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 (x)^9$$

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Machine Learning

CMPSCI 589

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Linear Regression

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